Bank Runs and Market Value Solvency

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Motivation: interest rate risk and the 2023 banking crisis

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  current regulation addresses the first two, through **capital requirements**

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Our goal:

1. Understand how **run risk** shapes demand for long bonds
2. Evaluate policies proposed to handle **interest rate risk**

This paper

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   - If depositors **runs**: short-term liability $\longrightarrow$ long-bonds are a bad hedge

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3. **Policy exercises:**

   Capital requirements: assign weights to long-bonds and uninsured deposits
   - In a 2022-like hike: Deposits involved in failures fall up to 21%
   - *Ex-ante*: cutting the hedge generates **welfare losses**

   Deposit insurance:
   - Extra coverage reduces runs and **increase welfare**

Literature

1. Empirical finance and interest rate risk[1]
   Flannery (1981), Choi et al. (1992), Hirtle (1997), Bednar and Elamin (2014), English et al. (2018), Gomez et al. (2021), Drechsler et al. (2021), Paul (2023), Choi et al. (2023), Haddad et al. (2023), Greenwald et al. (2024), Jiang et al. (2024), DeMarzo et al. (2025), Cipriani et al. (2024), Begenau et al. (2025)
   ⇒ **Non-monotonic relationship between portfolio weights and uninsured leverage.**
2. Theoretical literature
   Bank choices and run risk: Ennis and Keister (2003), Ennis and Keister (2006)
   Dynamic hedging and the deposit franchise: Di Tella and Kurlat (2021), Whited et al. (2021), Wang et al. (2022), Wang (2025), Bolton et al. (2025), Arfaoui and Uhlig (2025), Schneider (2026), Mertens et al. (2026), Abad et al. (2026), Fleckenstein et al. (2026), Drechsler et al. (2026), Amador and Bianchi (2024)
   ⇒ **Interaction of hedging demand and runs**
   Coordination runs and policy: Ennis and Keister (2009), De Groot (2021), Pancost and Robatto (2026), Ikeda and Matsumoto (2026)
   ⇒ **Deposit insurance and endogenous runs that respond to policy**
3. Interest-rate risk policies
   Begenau and Landvoigt (2022), Miao et al. (2023), Begenau et al. (2026), Varraso (2025)
   ⇒ **Hedging deposit spreads vs. runs + deposit insurance**

[1] Surveys and overviews include: Hanson et al. (2024), Metrick (2024), Vuillemey (2019).

Model

Environment and preferences

- Time is continuous, indexed by $t \in \mathbb{R}_+$
- Measure one of banks indexed by i, with preference $u(c_{it})$
- An outside investor with a SDF that prices all assets in the economy

$$\frac{dM_t}{M_t} = -r_t\,dt + \frac{\pi}{\sigma}\,dB_t$$

- A deposit insurance fund

Interest rate risk

- Short rates evolve stochastically according to:

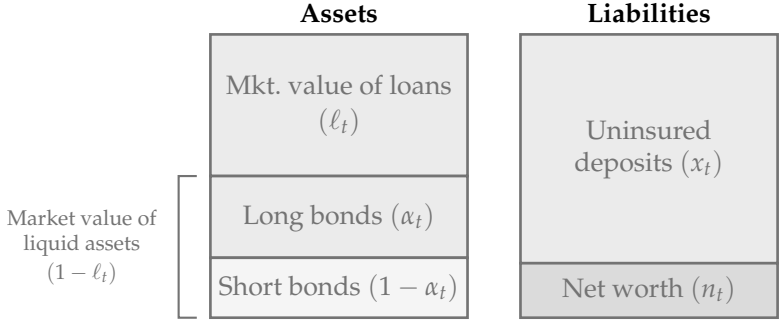
$$dr_t = \mu_r(r_t)dt + \sigma_r dB_t$$

- The short bond pays a return r_t

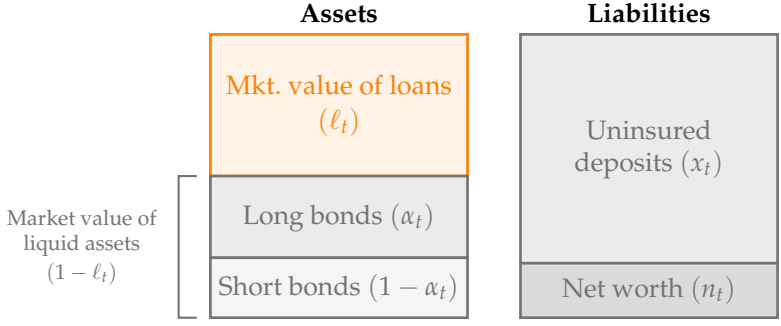
- The return on a **long maturity** with zero-coupon bond is:

$$\text{Long bond return}_t = \Big(r_t + \underbrace{\pi}_{\text{term premium}}\Big)dt - \underbrace{\sigma}_{\text{sensitivity to rate shocks}} dB_t$$

Balance Sheet: a bank i's individual state is (n_t, x_t, ℓ_t)



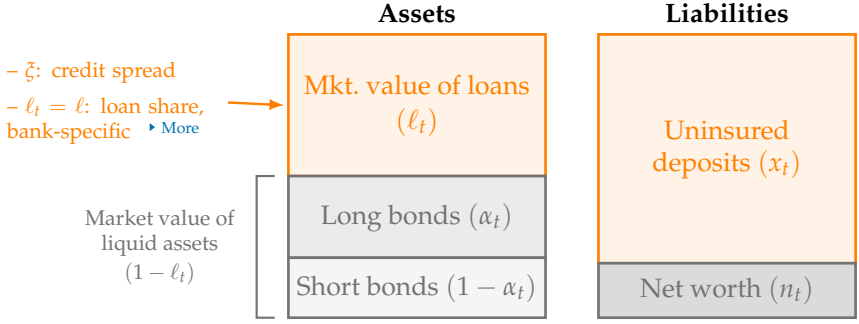
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– ξ: credit spread

– $\ell_t = \ell$: loan share, bank-specific ▸ More

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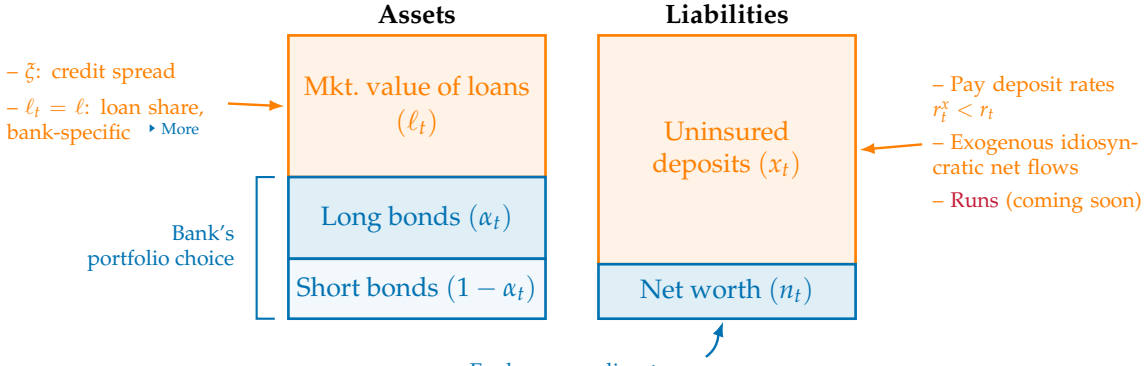


– Pay deposit rates $r_t^x < r_t$

– Exogenous idiosyncratic net flows

– Runs (coming soon)

Balance Sheet: a bank i's individual state is (n_t, x_t, ℓ_t)



$$dn_t = \Big[\big(r_t + \ell\xi + (1-\ell)\pi\alpha_t\big)(n_t + x_t) - c_t - x_t r_t^x - \kappa(n_t) - \tau_t\Big]dt$$
$$- \sigma\alpha_t(1-\ell)(n_t + x_t)\, dB_t$$

Deposits and runs

- A run for an active bank arrives with probability

$$\mathbb{P}_t(\text{run}) = \lambda(n_t, x_t, \ell)\, dt$$

In the paper: microfoundation with a coordination game among depositors

▸ Microfoundation

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$$\mathbb{P}_t(\text{run}) = \lambda\Big(\underbrace{x_t - [1-\ell+\ell\delta](n_t+x_t)}_{\text{uninsured deposits} - \text{liquidation value}} \Big)\, dt, \qquad \lambda' > 0$$

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- Define $T \equiv \inf\{t \geq 0 : \text{ Bank suffers a run}\}$

- Banks that suffer a run lose their **deposits and loans** Amador and Bianchi (2024)

$$\ell_t = x_t = 0, \qquad t \geq T$$

Bank's Problem

▸ HJB Equation

$$v(n, x, \ell, r, g) = \sup_{\{c_t, \alpha_t\}_{t \geq 0}} \mathbb{E}\left[\int_0^T e^{-\rho t} u(c_t) dt + e^{-\rho T} \overline{v}(n_T, r_T)\right]$$

state's law of motion:

Interest rate dynamics

Deposit dynamics and run intensity

Net worth dynamics

Distribution dynamics ▸ KF Equation

Deposit Insurance

▸ HJB ▸ Equilibrium

- Denote the distribution of active banks at time t by $g_t(n, x, \ell)$ ▸ KF Equation
- Banks are replenished according to the entrant distribution ψ_0
  (Constant mass of active banks)
- The deposit insurance fund balances its budget:

$$\tau_t \equiv \tau(r, g) = \zeta \times \int \lambda(n, x, \ell; \zeta) x g_t(n, x, \ell) \, dn dx d\ell \tag{1}$$

ζ: DIF-funded share of uninsured deposits

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- FDIC coverage ζ affects depositors' run incentives ▸ Microfoundation

Calibration and Validation

Functional forms

- Banker's utility:

$$u(c_{it}) = \frac{(Y + c_{it})^{1-\gamma} - Y^{1-\gamma}}{1-\gamma}$$

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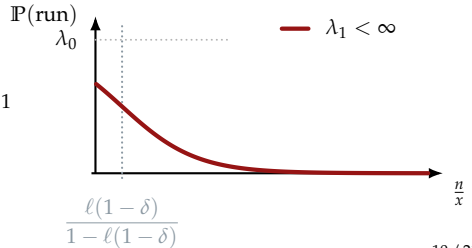
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- Run intensity:

$$\lambda(n, x, \ell) = \lambda_0 \left[1 + \exp\left\{\lambda_1 \left(\frac{n}{x} - \frac{\ell(1-\delta)}{1-\ell(1-\delta)}\right)\right\}\right]^{-1}$$



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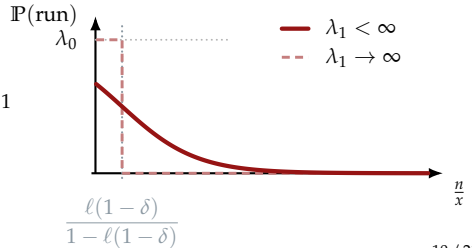
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Data and variables

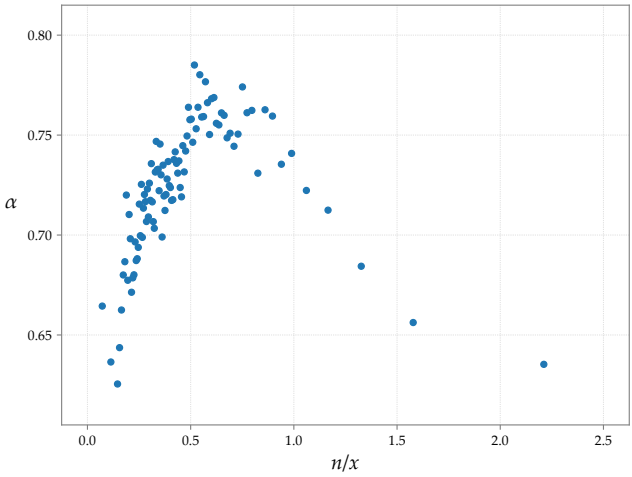
- Quarterly US Call Reports, 2010Q1–2025Q2
- Commercial banks with more than $1.5 billion in assets

$$\left(\frac{n}{x}\right)_{it} = \frac{\texttt{Book equity}_{it} + \texttt{Unrealized G/L}_{it}}{\texttt{Uninsured deposits}_{it}}$$

$$\alpha_{it} = \frac{\texttt{Securities}_{it} \texttt{ with maturity} > 3 \texttt{ months}}{\texttt{Liquid assets}_{it}}$$

Data: Long-bonds vs. balance-sheets

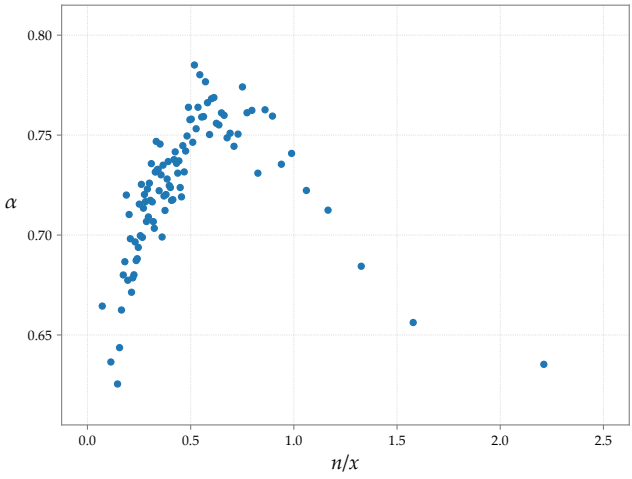
▸ More ▸ Some specific banks



Data: Long-bonds vs. balance-sheets

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$$\alpha_{it} = \beta_0 + \delta_t + \mu_i + \beta_1 \left(\frac{n}{x}\right)_{it} + \beta_2 \left(\frac{n}{x}\right)^2_{it}$$



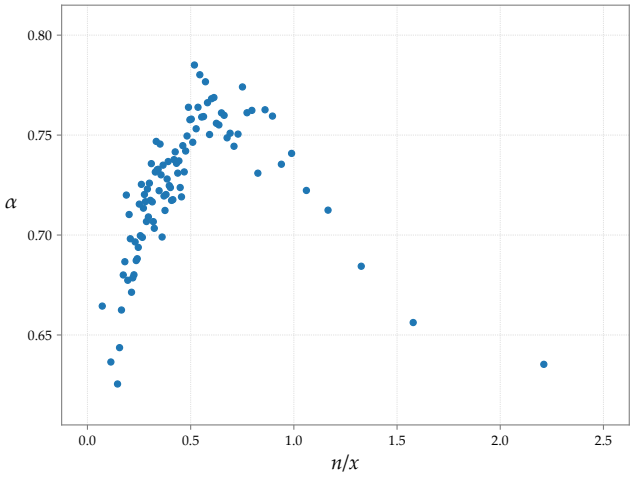
	Pooled	**Date FE**	**+ Bank FE**
Intercept	0.67		
$\left(\frac{n}{x}\right)_{it}$	0.17		
$\left(\frac{n}{x}\right)^2_{it}$	-0.08		

The **calibration targets**

Data: Long-bonds vs. balance-sheets

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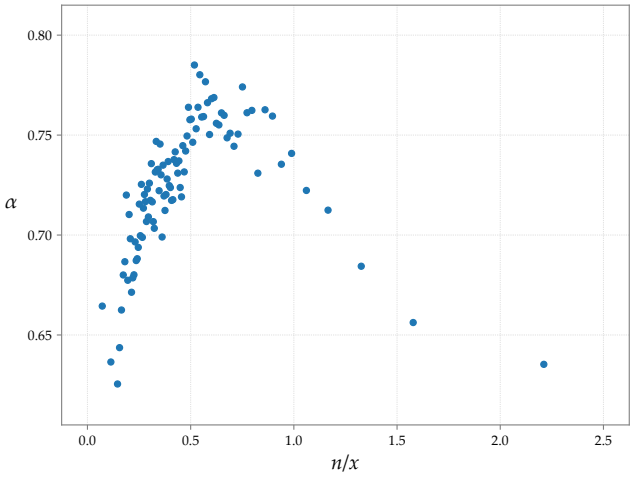
	Pooled	Date FE	+ Bank FE
Intercept	0.67	0.68	
$\left(\frac{n}{x}\right)_{it}$	0.17	0.14	
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Date FE: coefficients barely move

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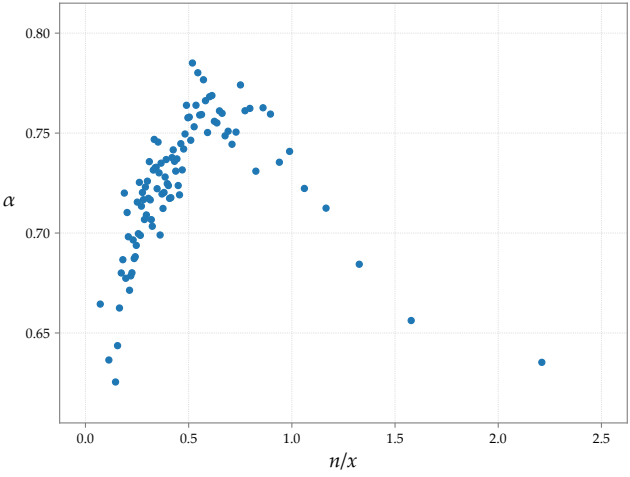


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Bank FE: coefficients halve

Calibration and model fit

▸ External parameters ▸ Alternative ▸ n/x distribution

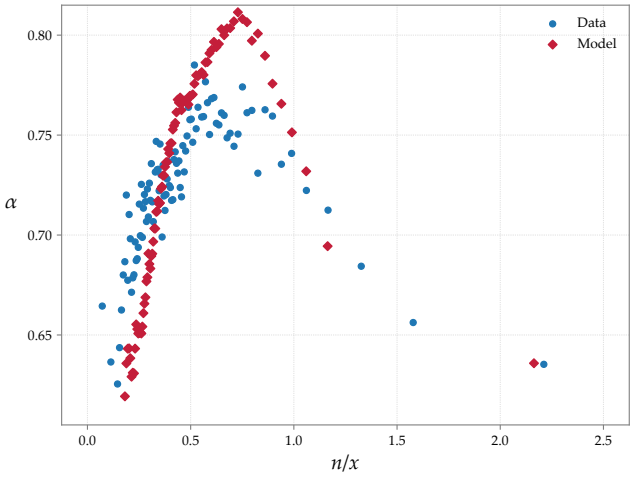


Parameter	Value	Description
ρ	0.03	Impatience rate
Y	0.34	Utility shift
δ	0.91	Loan sale price
λ_0	0.01	Level of run-risk
λ_1	6.78	Slope of run-risk
κ_1	0.01	Linear franchise cost

Moment	Data	Model
Targeted		
Avg. run rate	$< 0.36\%$	0.11%
Agg. c/n	0.07	0.08
Median n/x	0.42	0.44
Pooled: intercept	0.67	0.66
Pooled: $(n/x)_{it}$	0.17	0.19
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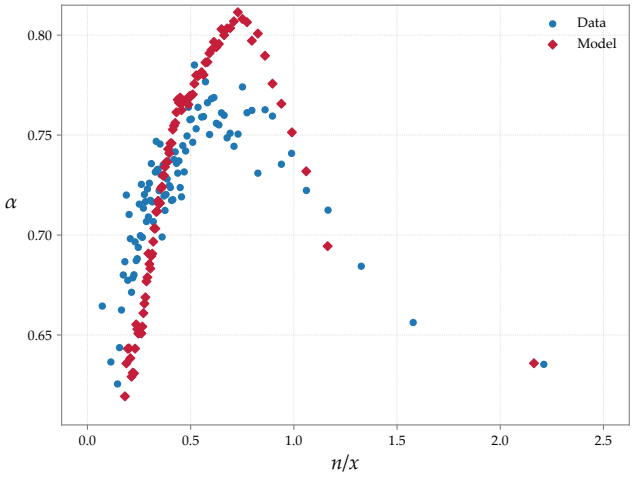
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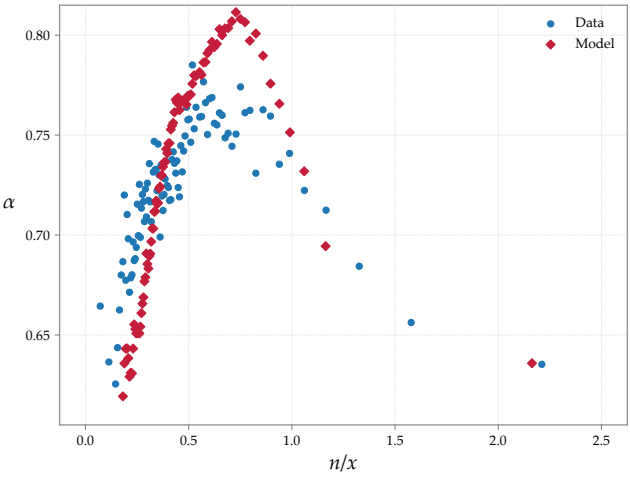
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Optimal Portfolio

Lemma *(Portfolio Characterization)*

The portfolio weight of long bonds is given by:

$$\alpha^*(n, x, \ell, r) = \textbf{Myopic demand} + \textbf{Dynamic hedging demand}$$

Optimal Portfolio: Myopic demand

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$$\alpha^*(n,x,\ell,r) = \underbrace{\text{Myopic demand}}_{\propto \pi/\sigma^2} + \underbrace{\text{Dynamic hedging demand}}_{\propto v_{nr}/v_{nn}}$$



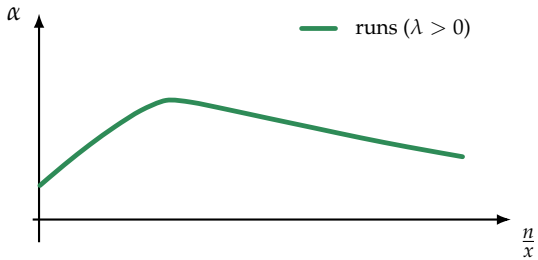
- With $\pi \approx 0$ the static demand is negligible ▸ More

Optimal Portfolio: **Dynamic hedging demand (with runs)**

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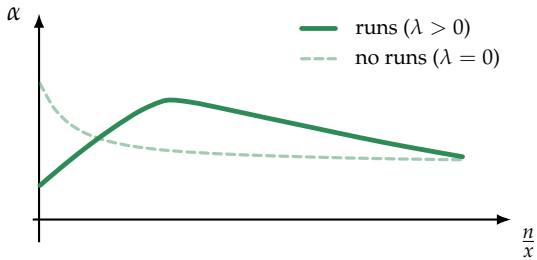


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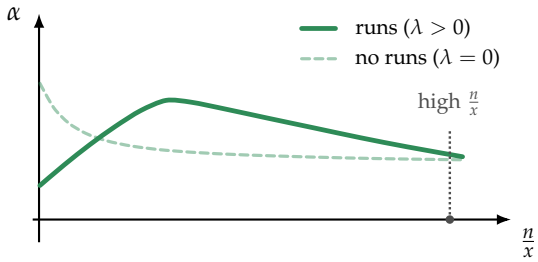
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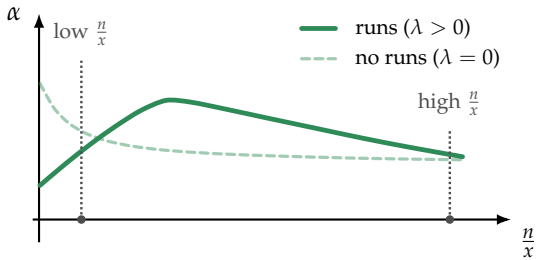
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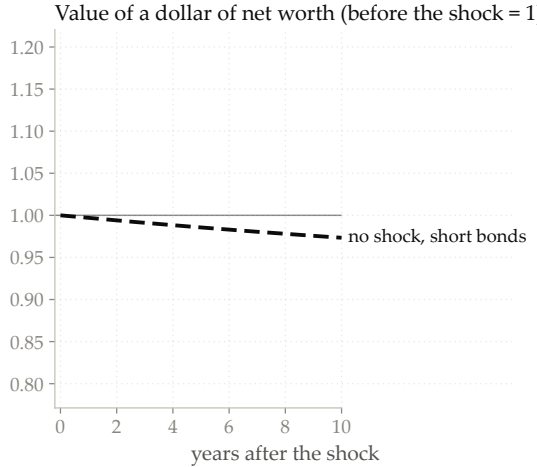
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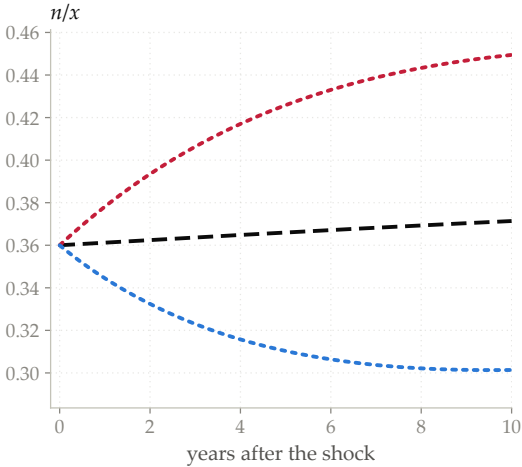
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- $\lambda \approx 0$: similar portfolios with/without runs
- High run probability banks want to preserve a scarce n

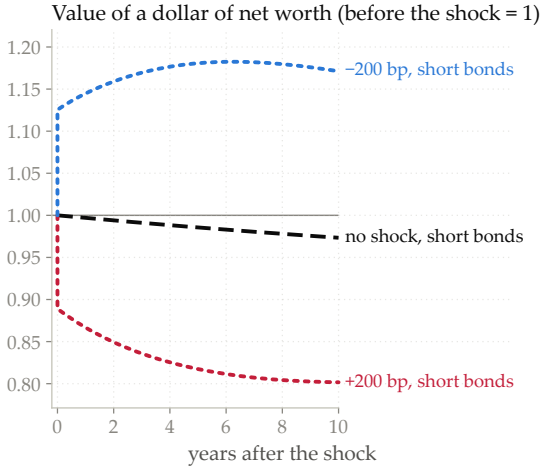
Evolution of n/x after larger shocks



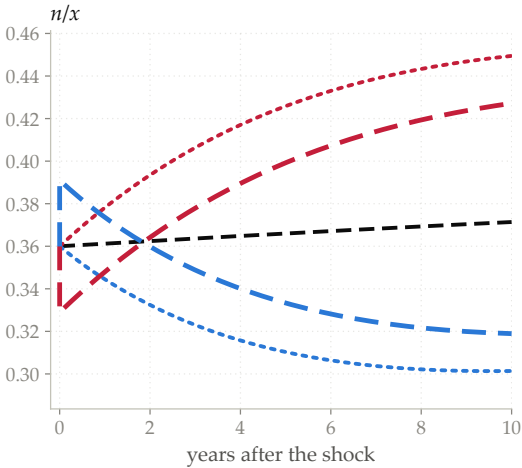


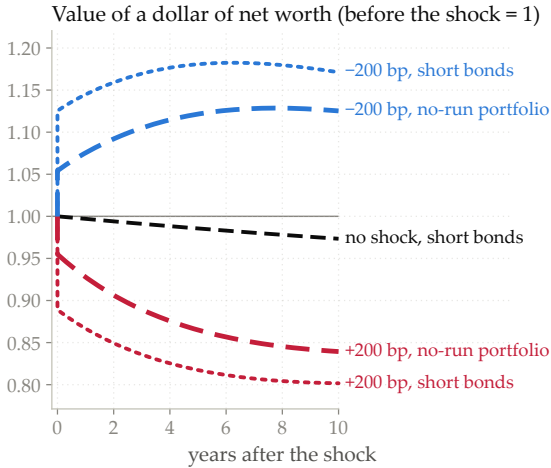
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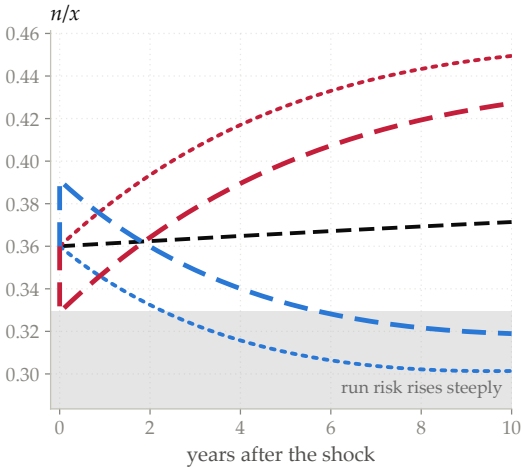


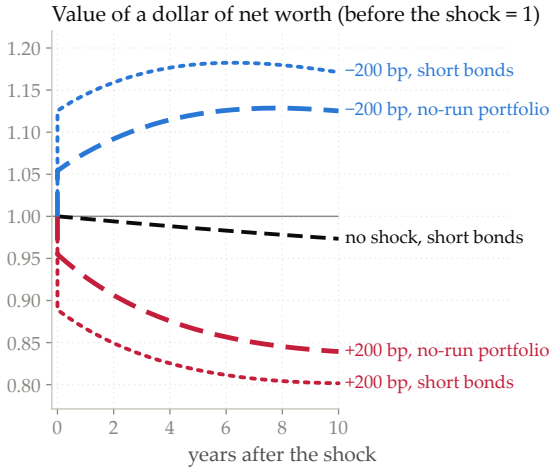
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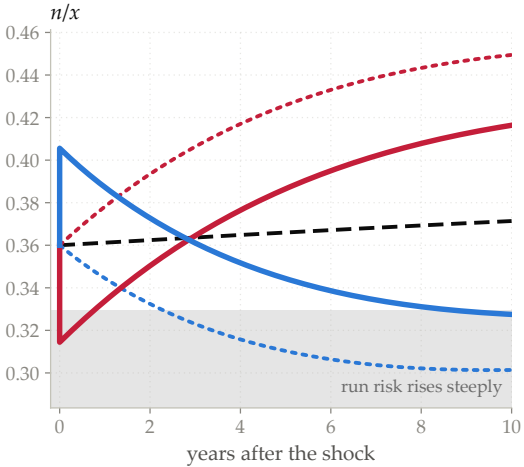


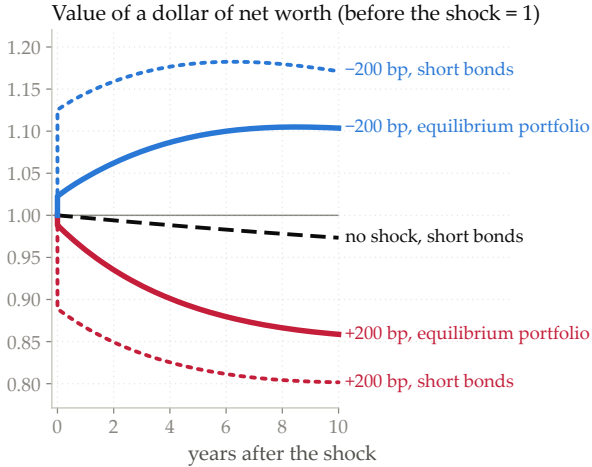
Evolution of n/x after larger shocks





Evolution of n/x after larger shocks





Validation with insured deposits

▸ More

- Insured depositors do not run: replace uninsured with insured deposits in n/x

$$\left(\frac{n}{x^{\text{Insured}}}\right)_{it} = \frac{\texttt{Book equity}_{it} + \texttt{Unrealized G/L}_{it}}{\texttt{Insured deposits}_{it}}$$

- Model: switch off runs ($\lambda_0 = 0$) and run the same regression on simulated banks

	Data *insured deposits*	**Model** *runs off* ($\lambda_0 = 0$)
Intercept	0.74	0.95
(n/x^{Insured})	−0.01	−0.62
$(n/x^{\text{Insured}})^2$	−0.27	−0.59

Validation with insured deposits

▸ More

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Without the run motive the hump disappears: α falls with n/x^{Insured}

Policy Exercises

Duration Restrictions

▸ Computing welfare

*At **any date**, the bank must absorb a shock Δr without net worth falling below ϕx*

$$n_t \geq (\Delta r \times \text{Duration}) \times \text{Long Bonds} + \phi \times \text{Un. Depositors}$$

$$\Delta r = +100 \text{ bp} \Rightarrow \text{weight on long bonds} \approx 7\%$$

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	Deposits in run banks	Sec. losses in 2022 (%)	DIF fee (bp)
FDIC, no floor	1.00	−17.7	1.12
FDIC + floor $\phi = 0.1$	0.99	−17.9	1.11
FDIC + floor $\phi = 0.2$	0.95	−17.1	1.07
FDIC + floor $\phi = 0.3$	0.84	−13.1	0.95
FDIC + floor $\phi = 0.4$	0.79	−7.7	0.89

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$\Rightarrow$ The hikes in 2022 were **a tail event**.

Welfare gains of the duration restrictions

▸ Computing welfare

	Bankers, CE (bp of div.)
FDIC, no floor	0
FDIC, floor $\phi = 0.1$	-5.5
FDIC, floor $\phi = 0.2$	-15.9
FDIC, floor $\phi = 0.3$	-34.3
FDIC, floor $\phi = 0.4$	-62.2

Ex-ante, restrictions on the long-bond holdings reduce welfare

Welfare gains of FDIC coverage

▸ Computing welfare

- Higher ζ: fewer runs ($\lambda(n, x, \ell; \zeta)$ weakly decreasing on ζ)
- Higher ζ: higher taxes on banks

	Bankers, CE (bp of div.)
No FDIC ($\zeta = 0.00$)	0
FDIC ($\zeta = 0.035$)	+33.9

Conclusion

1. **How run risk shapes demand for long bonds**
   - Runs hit banks with low n/x: their deposits look like short-term liabilities
   - Hedging motive is **strongest at intermediate** n/x

2. **Policies proposed to handle interest rate risk**
   - Positive risk-weights to long bonds and un. deposits

     $\downarrow$ failures, but removes hedging $\Rightarrow$ $\downarrow$ welfare
   - FDIC coverage

     $\downarrow$ incentives to run $\Rightarrow$ $\downarrow$ inefficiencies from coordination failures

Thank you!

Appendix

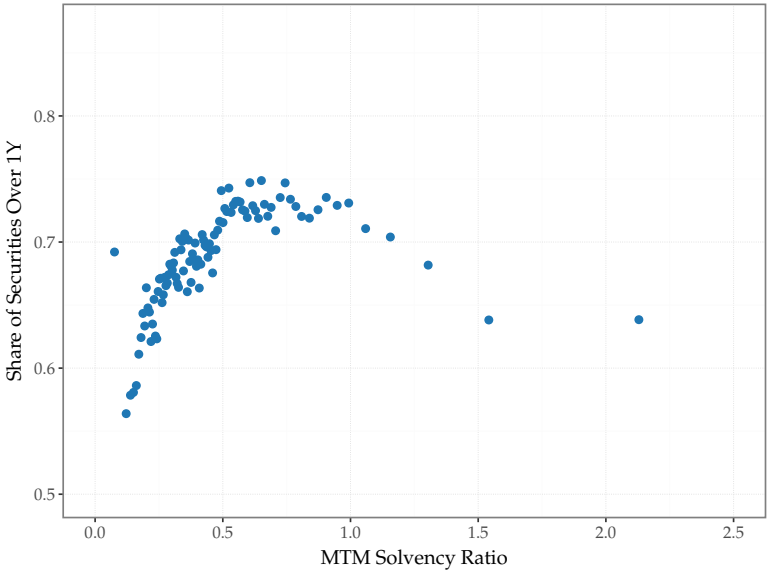
Data Appendix

Back to hump-shape

1. Detailed by size bins More
2. Account for loan share More
3. Market value of loans More
4. What is in each bin? More
5. Fixed Effects More
6. Account for the intensive margin duration choice More
7. BHC aggregation More
8. Insured deposits in the solvency ratio More

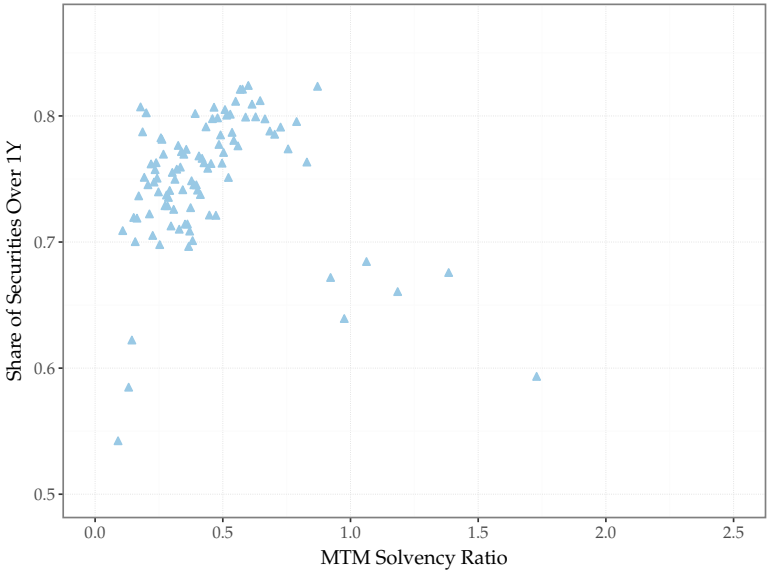
1.5B-10B in assets

Back



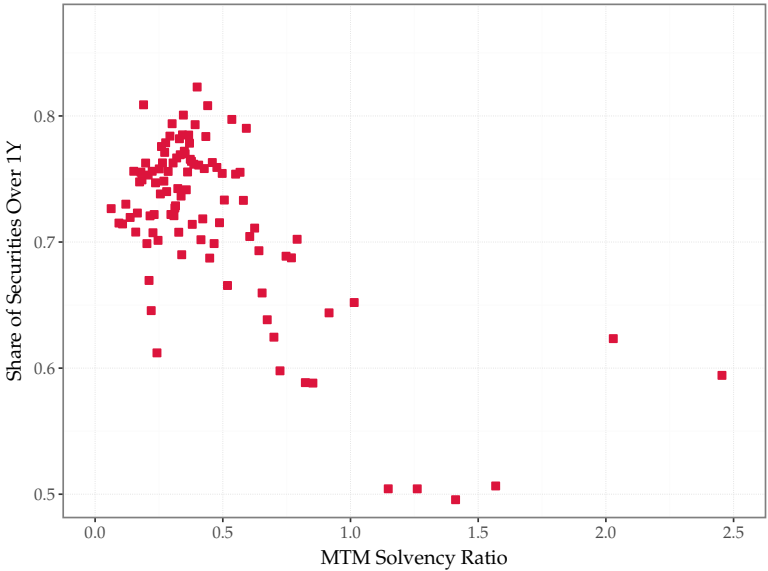
10B-50B in assets

Back



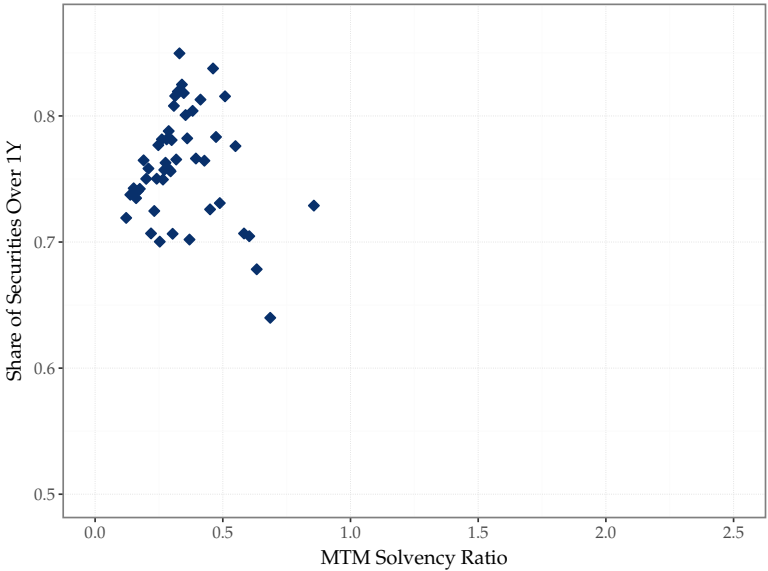
50B-250B in assets

Back



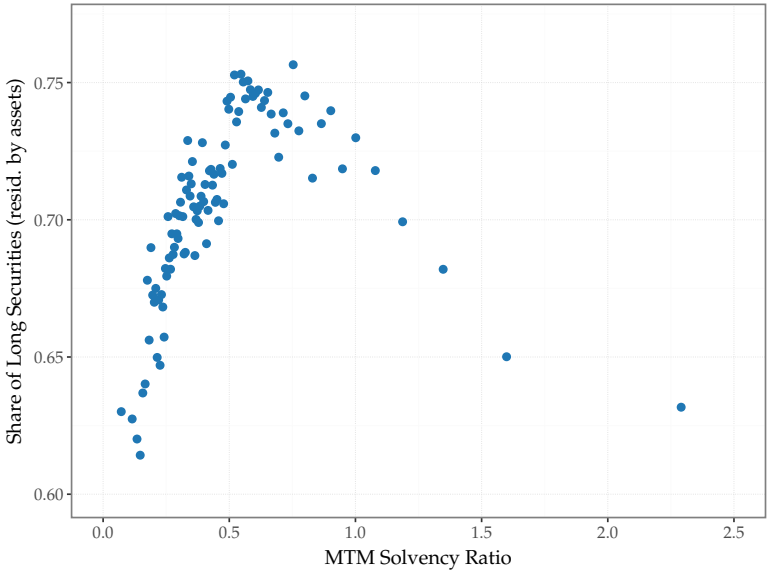
>250B in assets

◂ Back



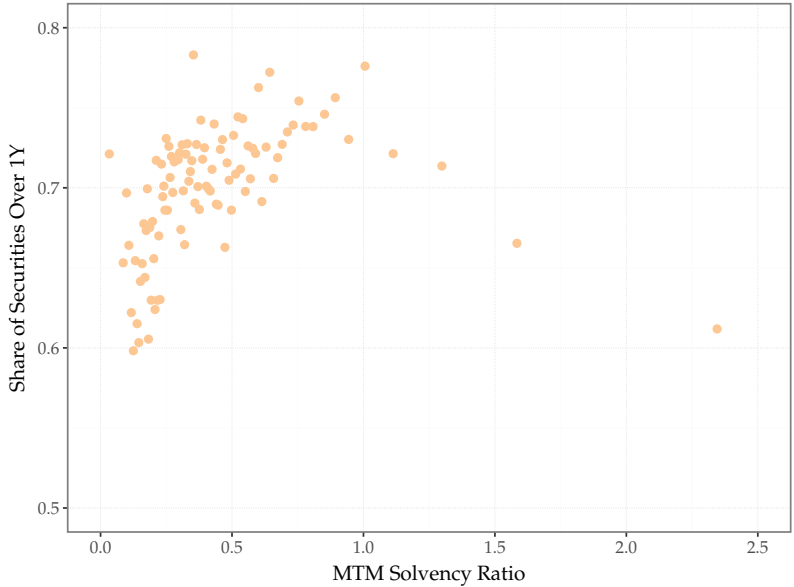
Residualize by bank's assets

Back



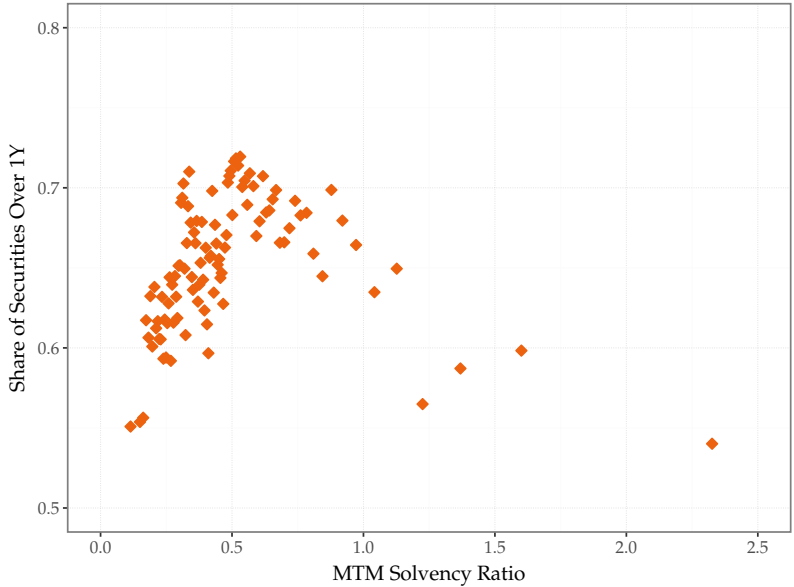
1st quartile of loan share

Back



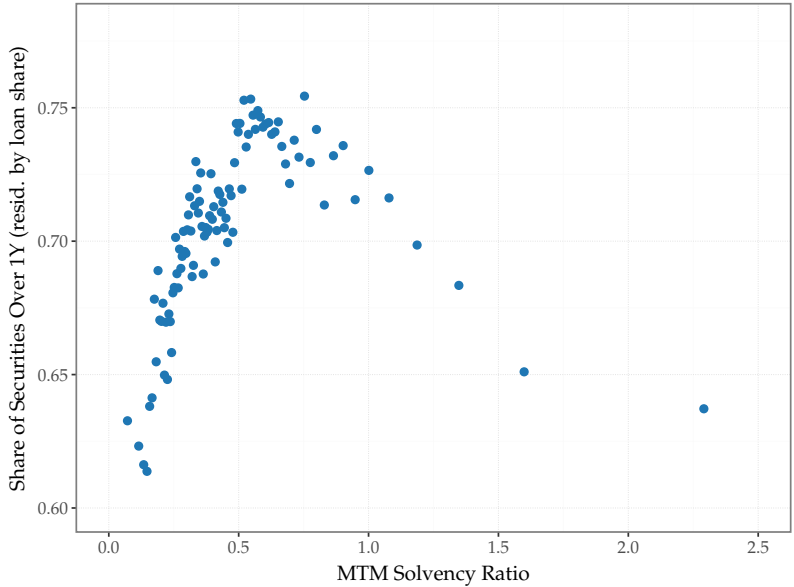
4th quartile of loan share

Back



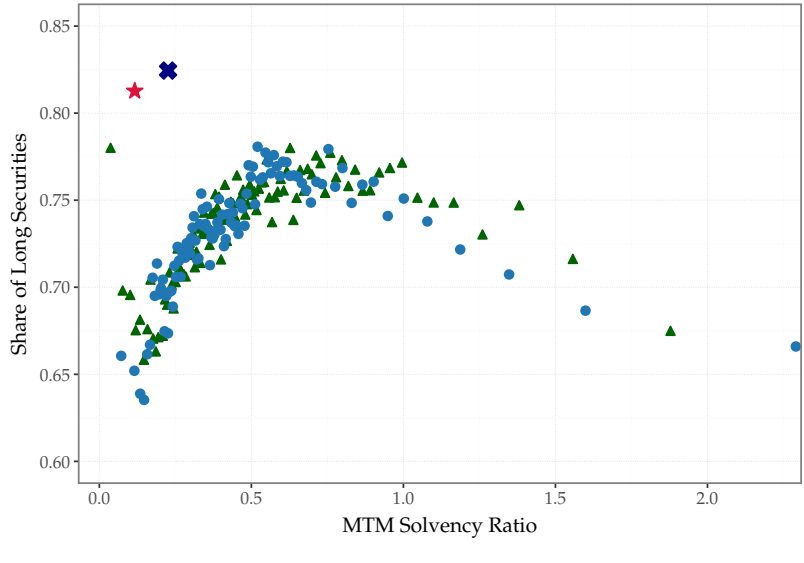
Residualize for loan share

Back



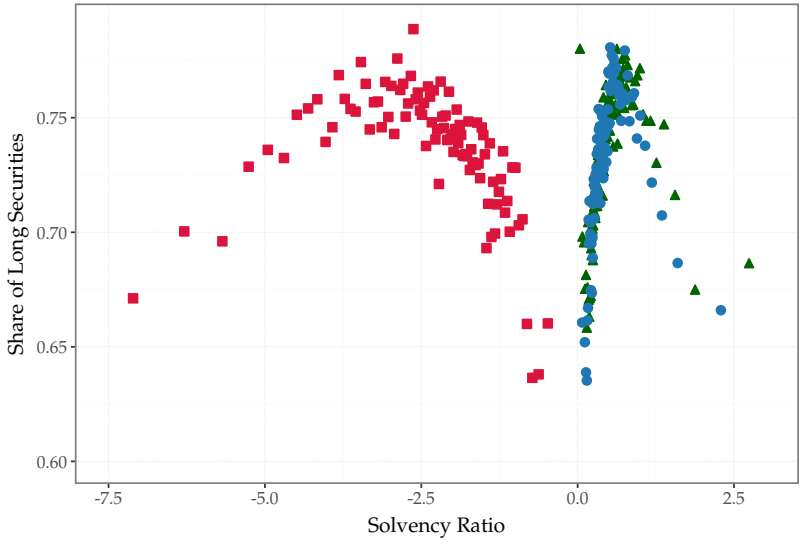
Market value of loans

Back



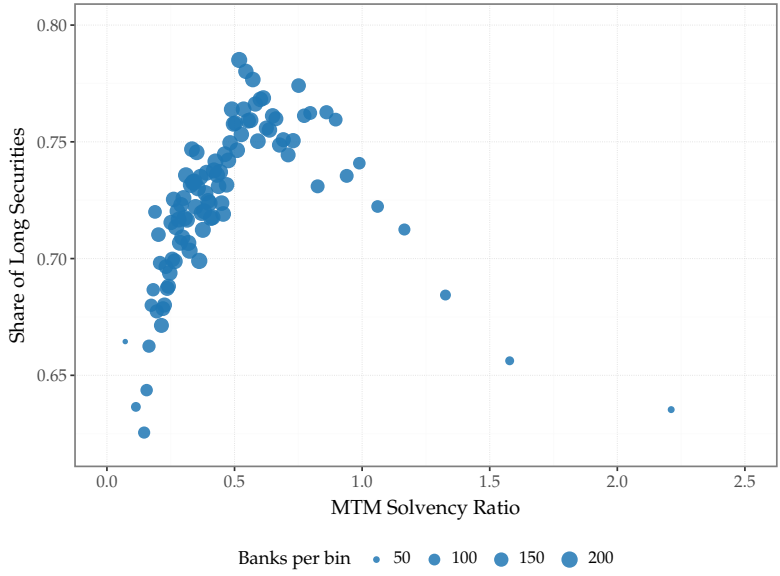
Excluding face value of loans

Back



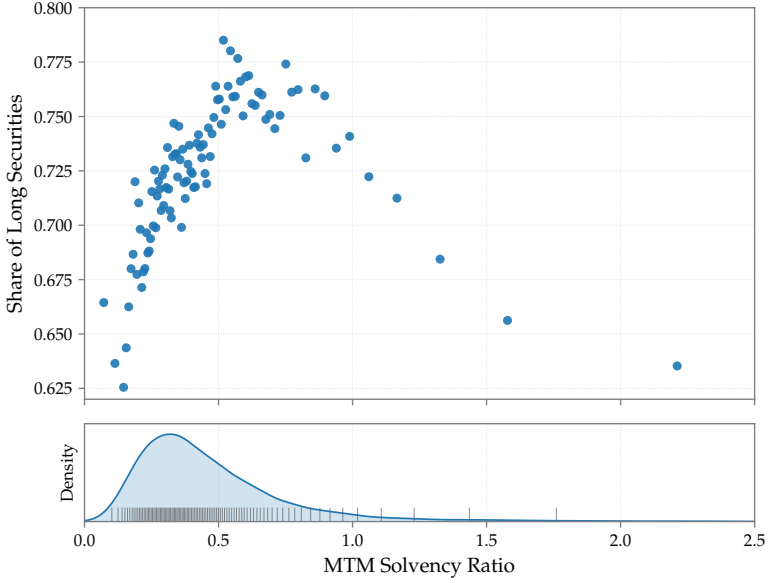
How important is a bin?

Back



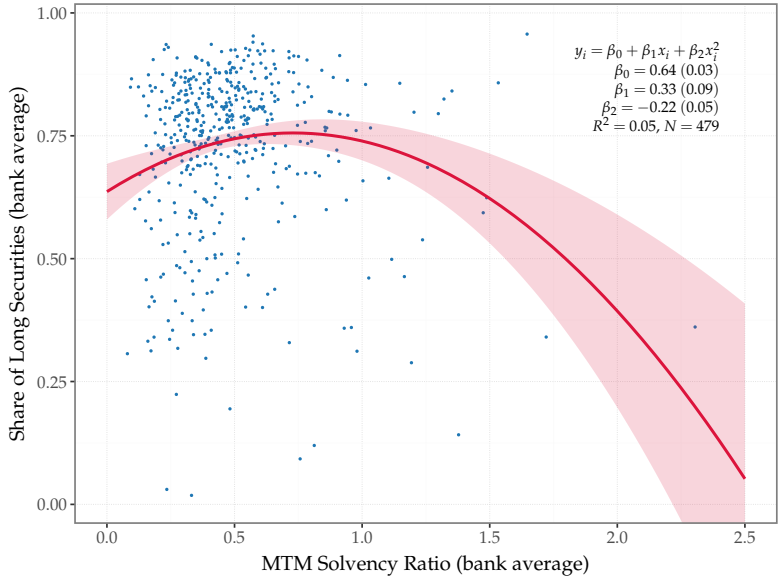
How important is a bin?

Back



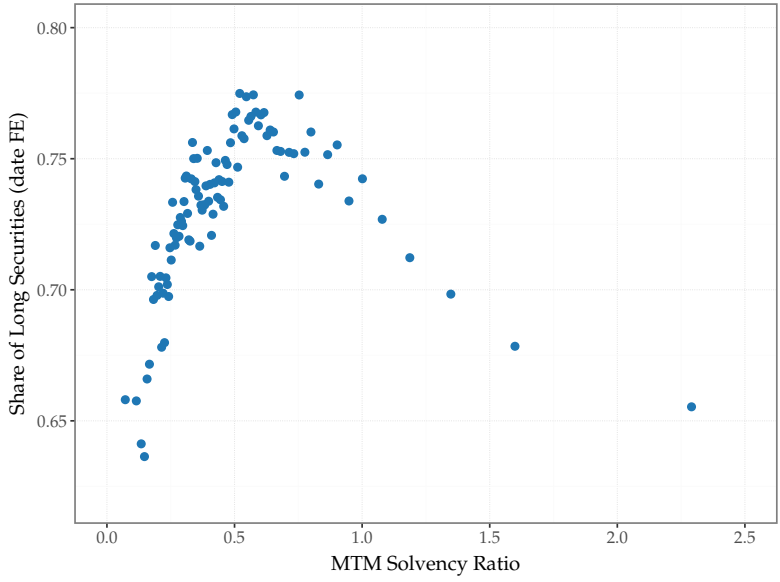
How important is a bin?

Back



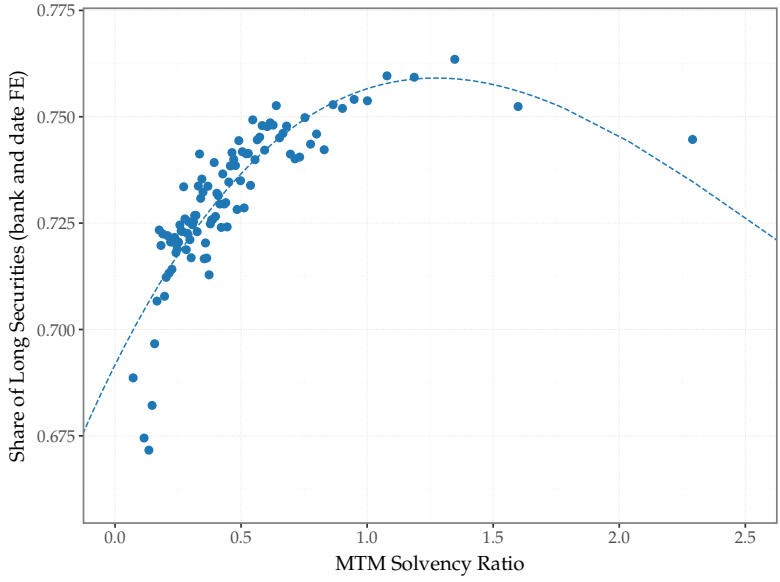
Time fixed-effects

Back



Time + bank fixed-effects

Back



$\text{Share of Long Securities}_{it} \times (\text{Maturity} \mid > 1y)_{it}$

Back

Average maturity among securities longer than one year (in years):

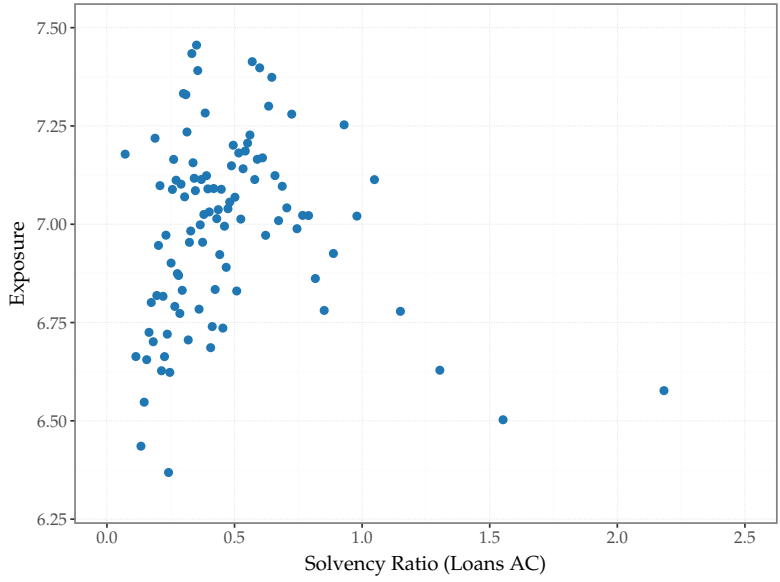
$$(\texttt{Maturity} \mid > 1y)_{it} = \frac{2\,\texttt{Sec}_{it}^{1\text{–}3y} + 4\,\texttt{Sec}_{it}^{3\text{–}5y} + 10\,\texttt{Sec}_{it}^{5\text{–}15y} + 15\,\texttt{Sec}_{it}^{>15y}}{\texttt{Sec}_{it}^{1\text{–}3y} + \texttt{Sec}_{it}^{3\text{–}5y} + \texttt{Sec}_{it}^{5\text{–}15y} + \texttt{Sec}_{it}^{>15y}}$$

The y-axis combines the extensive and intensive margins of the duration choice:

$$\texttt{Exposure}_{it} = \texttt{Share of Long Securities}_{it} \times (\texttt{Maturity} \mid > 1y)_{it}$$

Share of Long Securities$_{it}$ $\times$ (Maturity $| > 1y)_{it}$

Back



Asset-side IR exposure: time fixed-effects

◂ Back

From returns to interest-rate loadings [following Begenau et al. (2025)]

1. Quarterly holding-period returns of Treasuries, AAA/BBB corporates, and MBS, by maturity m
2. Two-factor regression on the 5y default-free and 5y BBB returns:

$$r_{mt} = \alpha_m + \beta_m^{DF} r_t^{DF} + \beta_m^{BBB} r_t^{BBB} + u_{mt}$$

3. The BBB return itself carries rate risk: $r_t^{BBB} = a\, r_t^{DF} + (1-a) f_t^{credit}$, so the interest-rate loading of maturity m is

$$b_m^{IR} = \beta_m^{DF} + a\, \beta_m^{BBB}$$

(a: OLS of r^{BBB} on r^{DF}, excluding stress quarters)

Asset-side IR exposure: time fixed-effects

◂ Back

From loadings to a bank-level exposure

5. Each Call Report maturity bucket valued at a representative maturity:
   $\leq$3m $\to$ 0, 3m–1y $\to$ 0.625, 1–3y $\to$ 2, 3–5y $\to$ 4, 5–15y $\to$ 10, $>$15y $\to$ 15 years
6. Loading curve by asset class: Treasuries $\to$ Treasury curve; MBS $\to$ AAA; other securities and loans $\to$ AAA/BBB midpoint ("safe")
7. Dollar exposure: sum bucket positions times loadings over securities and loans,

$$\texttt{IR\$}_{it} = \sum_{m} \texttt{Position}_{itm} \times b_m^{IR}$$

   (security buckets rescaled to the fair-value total)
8. y-axis: normalize by total assets, $\texttt{IR Exposure}_{it} = \texttt{IR\$}_{it} / \texttt{Total Assets}_{it}$

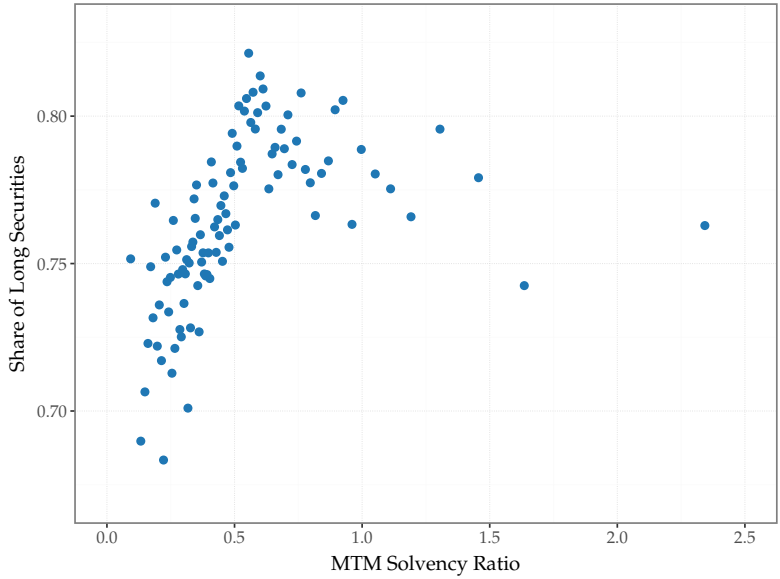
Asset-side IR exposure: time fixed-effects

Back



BHC aggregation

Back



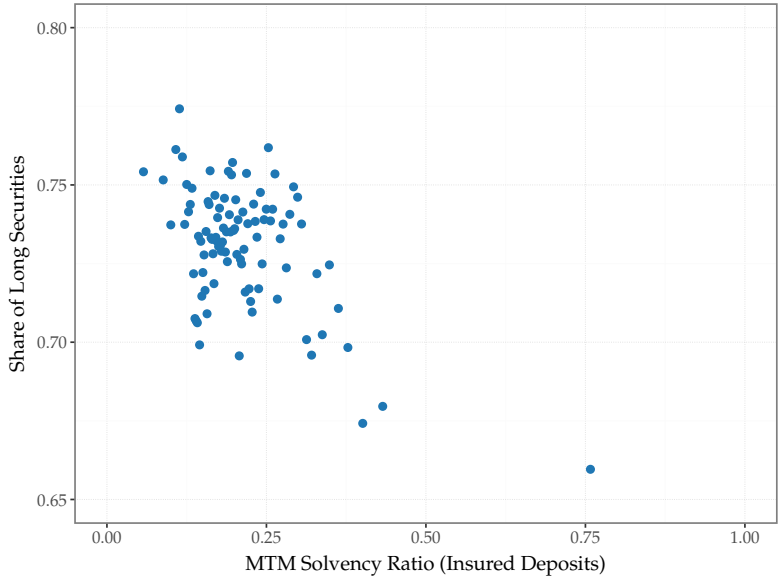
Insured deposits: no fixed effects

Back



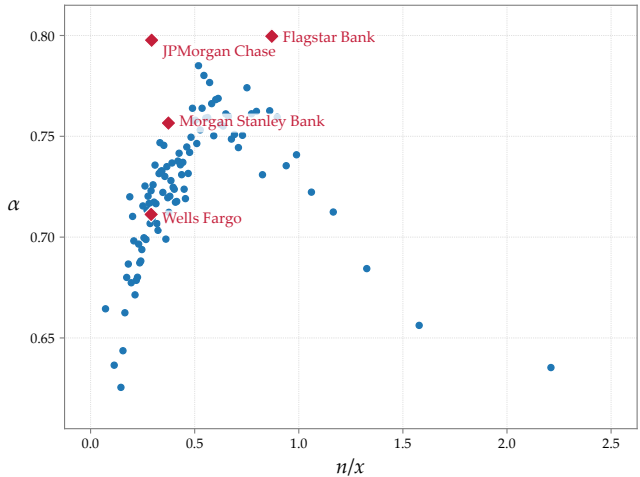
Insured deposits: time fixed-effects

Back



Some specific banks

◂ Back to hump-shape



Diamonds: each bank's average over its quarters in the sample (2010–2025). Flagstar Bank filed as New York Community Bank until 2022Q3.

More sensitive spreads, more long bonds

▸ More ◂ Back

Define the deposit spread:

$$\texttt{Deposit Spread}_{it} = \texttt{FedFunds Rate}_{t} - \texttt{Deposit Rate}_{it}$$

More sensitive spreads, more long bonds

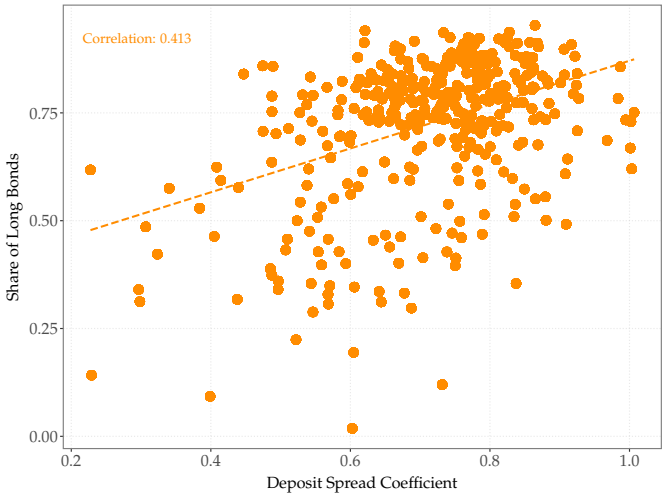
▸ More ◂ Back

Following DeMarzo et al. (2025), we estimate the following regression:

$$\texttt{Deposit Spread}_{it} = \mu_i + \underbrace{(1-\beta_i)}_{\text{spread sensitivity}} \times \texttt{FedFunds Rate}_t + \epsilon_{it}$$

More sensitive spreads, more long bonds

▸ More ◂ Back



Robustness: deposit-spread sensitivity

Back

Every estimate is bank by bank, with Newey–West standard errors. The dependent variable is always the deposit spread s_{it}^D, so each coefficient is $\partial s_{it}^D / \partial r_t$, not a deposit-rate pass-through.

- **Levels, contemporaneous** (baseline): $s_{it}^D = \alpha_i + \beta_i^{L0}\, r_t + \varepsilon_{it}$
- **Differences, contemporaneous**: $\Delta s_{it}^D = a_i + \beta_i^{D0}\, \Delta r_t + u_{it}$
- **Levels, cumulative** over 3 lags:

$$s_{it}^D = a_i + \sum_{\ell=0}^{3} \beta_{i,\ell}^L\, r_{t-\ell} + u_{it}, \qquad \beta_i^{L,\text{cum}} = \sum_{\ell=0}^{3} \beta_{i,\ell}^L$$

- **Differences, cumulative** over 3 lags:

$$\Delta s_{it}^D = a_i + \sum_{\ell=0}^{3} \beta_{i,\ell}^D\, \Delta r_{t-\ell} + u_{it}, \qquad \beta_i^{D,\text{cum}} = \sum_{\ell=0}^{3} \beta_{i,\ell}^D$$

Robustness: identified monetary-policy surprises

Back

m_t: Bauer & Swanson (2023) high-frequency surprises, orthogonalized to pre-announcement macro news, summed within the quarter.

- **MP surprise, cumulative** over 3 lags:

$$\Delta s^D_{it} = a_i + \sum_{\ell=0}^{3} \theta_{i,\ell}\, m_{t-\ell} + u_{it}, \qquad \theta^{\text{cum}}_i = \sum_{\ell=0}^{3} \theta_{i,\ell}$$

- **Local projection** (Jordà, 2005) at horizon $h = 4$:

$$s^D_{i,t+h} - s^D_{i,t-1} = a_{i,h} + \psi_i(h)\, m_t + \Gamma'_{i,h} X_{i,t-1} + u_{i,t+h}$$

with $X_{i,t-1}$: two lags of Δs^D_{it} and of Δr_t, all strictly lagged

- **Local projection, IV**: the rate itself is the regressor, instrumented by m_t

$$r^x_{i,t+h} - r^x_{i,t-1} = \delta^h_0 + \delta^h_1 \underbrace{(r_{t+h} - r_{t-1})}_{\text{instrumented by } m_t} + \Gamma' X_{i,t-1} + \lambda_i + v_{i,t+h}$$

r^x: deposit rate, i: 1-year Treasury; bank FE.

The first two are per 100 bp of the 1-year Treasury; δ^h_1 is the deposit beta, so $\partial s^D / \partial r = 1 - \delta^h_1$.

Robustness: sensitivity and long securities

Back

Estimator	Banks	Mean	Corr. with share long sec.
Levels, contemporaneous	379	0.71	0.43
Differences, contemporaneous	369	0.92	0.32
Levels, cumulative over 3 lags	377	0.63	0.43
Differences, cumulative over 3 lags	369	0.69	0.34
IV-Local projection, horizon 4 (cumulative)	321	0.64	0.14

The last row is in bp of spread per 100 bp impact rise in the 1-year Treasury, so their means are on a different scale; correlations are scale-free.

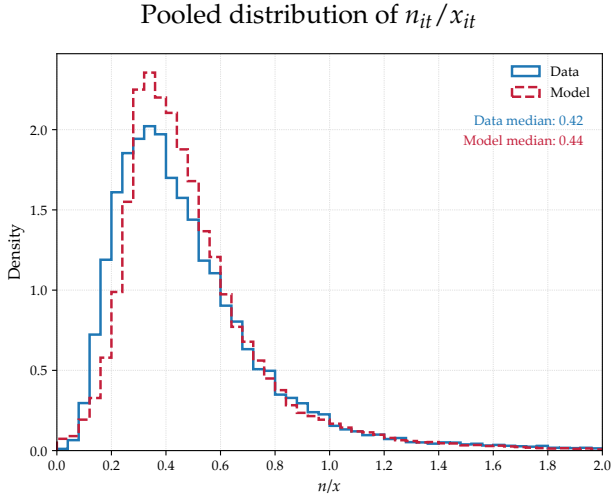
Externally calibrated parameters

Back

Parameter	Value	Description	Source
$\bar{x},\ \sigma_x$	0.00	Constant deposits	Baseline
k	0.01	Withdrawal cost	Baseline
ζ	**0.035**	**DIF coverage ratio**	Baseline More
γ	2.00	Risk aversion	Standard
$\bar{r}$	0.02	Average rate	Data More
θ_r	0.14	Mean reversion of rates	Data More
σ_r	0.01	Volatility of rates	Data More
ς	-0.01	Price of risk	Data More
ι	0.03	Inflation rate	Data
ξ	0.03	Credit spread	Data More
β	0.30	Deposit rate sensitivity	Drechsler et al. (2026)

Distribution of n/x: model vs. data

Back



DIF-funded share

Back

- At failure the DIF pays ζ per uninsured dollar; depositors lose $[\ell(1-\delta)-\zeta]x$, so $0 \le \zeta < \ell(1-\delta)$

- FDIC failures 2010–2024: 99.7% of uninsured deposits at failed banks were in whole-deposit P&As

  uninsured depositors are made whole, so the DIF bears the full loss: ζ at its upper bound $\ell(1-\delta)$

- With $\delta = 0.91$, $\ell(1-\delta) = 0.035$ at $\ell = 0.39$, the 5th percentile of the loan share (2010–2024)

$$\zeta = 0.035$$

The net DIF cost per uninsured dollar at SVB and Signature is higher, but it also covers insolvency losses that the model leaves with the banker

Alternative: calibration with real rates

◂ Back ▸ Model fit

Parameter	Value	Description
Externally calibrated (3m SPF real rate block; all else as in the main calibration)		
$\bar{r}$	-0.005	Average rate
θ_r	0.27	Mean reversion of rates
σ_r	0.01	Volatility of rates
ς	-0.01	Price of risk
Internally calibrated		
γ	2.23	Risk aversion
Y	0.89	Utility shift
δ	0.83	Fire-sale cost
λ_1	6.78	Slope of run-risk
λ_0	0.011	Level of run-risk

Alternative: model fit with real rates

◂ Back ▸ Calibration

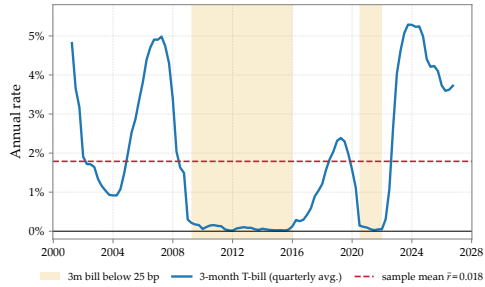
- Same model and targets, re-estimated with the 3m SPF **real** rate block

Moment	**Data**	**Model**
Targeted		
Avg. run rate	0.35%	0.38%
Table (1), pooled: Intercept	0.67	0.54
Table (1), pooled: MTM solvency ratio	0.17	0.12
Table (1), pooled: MTM solvency ratio^2	−0.08	−0.10
Untargeted		
Table (1), Date FE: MTM solvency ratio	0.14	0.11
Table (1), Date FE: MTM solvency ratio^2	−0.07	−0.10
Table (1), Date + bank FE: MTM solvency ratio	0.08	−0.04
Table (1), Date + bank FE: MTM solvency ratio^2	−0.03	−0.03

Short rate process: data and estimation

◂ Back ▸ Lower bound

- **Data:** 3-month T-bill (FRED `TB3MS`), quarterly average of monthly rates, 2001Q1–2026Q3 ($T = 103$); the 1-year CMT as a check
- **Exact discretization** of $dr = \theta_r(\bar{r} - r)dt + \sigma_r dB$ at $\Delta = 1/4$: $r_{q+1} = a + b\, r_q + u_{q+1}$ with $b = e^{-\theta_r \Delta}$ and $\mathrm{Var}(u) = \sigma_r^2(1 - b^2)/(2\theta_r)$
- **OLS:** $\hat{b} = 0.964$ (0.023), $R^2 = 0.95$; $\bar{r}$ imposed at the sample mean (the free intercept divides two small numbers)
- **Mapping:** $\theta_r = -4 \log \hat{b} = 0.145$, $\sigma_r = 0.0084$, $\bar{r} = 0.0179$
- **Implied:** half-life 4.8 years; quarterly innovation s.d. 41 bp; $\sigma_r/\theta_r = 0.058$ (long-bond exposure per unit shock)



Rounded to $\bar{r} = 0.02$, $\theta_r = 0.14$, $\sigma_r = 0.01$ in the table.

Short rate process: the lower bound

Back

- 35 of 103 quarters (34%) have the 3-month bill below 25 bp (2009–2015, 2020–2021)
- A linear Gaussian OU has no floor: flat ELB stretches read as persistence, so $\hat{b} \to 1$ and θ_r, σ_r are biased down
- Re-estimate on pairs with *both* quarters off the bound ($T = 65$):

	Full sample	Off the bound
θ_r (mean reversion)	0.146	0.292
σ_r (instantaneous vol.)	0.0084	0.0103
Half-life (years)	4.8	2.4
Quarterly innovation s.d. (bp)	41	50
σ_r/θ_r (long-bond exposure)	0.058	0.035

- The 95% interval for $\hat{b}$ includes 1 in both columns: $1/\theta_r$ is bounded below (3.0 and 1.6 years) but not above
- Residuals are not white (AC(1) $= 0.71$, Ljung–Box rejects): the quarterly innovation and σ_r/θ_r are the robust objects
- The calibration uses the full-sample column; the off-bound column brackets it

Deposit process

Back More

- **Sample:** 921 banks, 28,252 bank-quarters, 2010Q1–2025Q2; 27,155 consecutive-quarter changes

- **De-mean** log uninsured deposits by bank and quarter fixed effects (iterative demeaning):

$$\log D_{it} = \mu_i + \delta_t + z_{it}, \qquad x_{it} \equiv 100\, e^{z_{it}}$$

  x: mean 104, median 100; 1st–99th percentiles 45–209

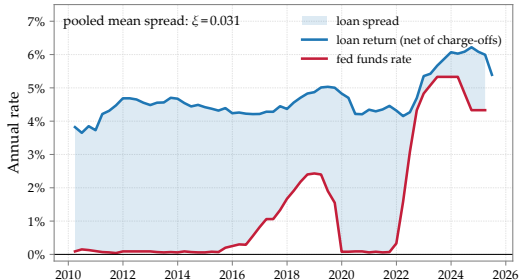
- **Exact discretization** of $dx = \bar{x}\, dt + \sigma_x\, dW^x$ at $\Delta = 1/4$: $x_{t+\Delta} - x_t = \bar{x}\Delta + \sigma_x \sqrt{\Delta}\, \varepsilon_t$

- **Estimates** (quarterly Δx winsorized at the 1st/99th percentiles, -31.8 and $+39.2$):
  - mean $\Delta x = 0.13$ per quarter $\Rightarrow \bar{x} = 0.53$ per year, set to 0
  - s.d. $\Delta x = 10.2$ per quarter $\Rightarrow \sigma_x = 10.2/\sqrt{1/4} = 20.5$
  - without winsorizing: s.d. 15.2 $\Rightarrow \sigma_x = 30.5$ (fat tails, next slide)

Loan block

Back

- **Loan return** (annualized, net of credit losses): $4\times$(interest income on loans and leases $-$ charge-offs $+$ recoveries) / total loans
- **Loan spread:** loan return $-$ fed funds rate
- ξ: pooled mean over the analysis sample (28,215 bank-quarters): 0.031; median 0.034; loan-weighted 0.030
- The spread moves with the policy rate (4.6% in 2020Q1, 0.7% in 2024Q1); the model holds ξ at its mean



Quarterly cross-sectional means.

Price of risk

◂ Back ▸ Term Premium

- The long bond's expected excess return is $\pi \equiv \varsigma\sigma_r/\theta_r$ (Lemma), so ς is set to match the average term premium:

$$\varsigma = TP \times \frac{\theta_r}{\sigma_r}$$

- *TP*: ACM 5-year term premium (Adrian, Crump and Moench), end of quarter, 2010Q1–2025Q4: mean -0.04%

- $\varsigma = -0.0004 \times 0.145/0.0084 = -0.0069 \simeq -0.01 \Rightarrow \pi \approx -4$ bp: the myopic demand is negligible

- Over 2001–2025 the 5-year premium averages $+0.32\%$ ($\varsigma \approx +0.055$); the post-2010 window matches the Call Report sample

Franchise cost parameter

Back

- **Operating cost** from the Call Report, annualized: $C_{it} = 4\times$(noninterest expenses $-$ service charges on deposits $-$ other noninterest income $-$ credit-card interest income)

- **Model units:** divide by the bank's deposit trend $e^{\mu_i+\delta_t}$ from the deposit block, as for n and x: $\kappa_{it} = 100\, C_{it}/\text{trend}_{it}$, $n_{it} = 100\, N_{it}/\text{trend}_{it}$

- **Specification** $\kappa(n) = \frac{\kappa_0}{2} n^2 \Rightarrow \kappa/n = \frac{\kappa_0}{2} n$: pooled OLS through the origin on 27,689 bank-quarters (913 banks)

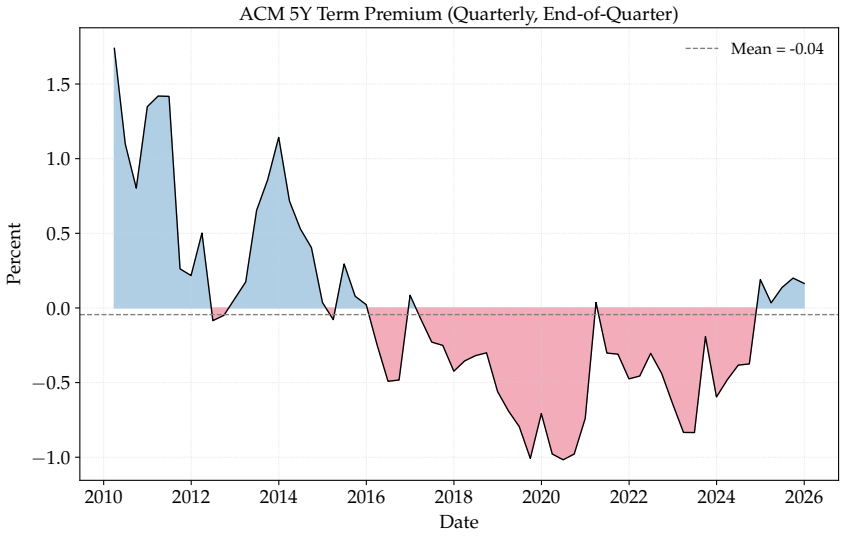
  slope 0.0027 (s.e. 0.00003) $\Rightarrow \kappa_0 = 0.0054 \simeq 0.005$

- Median $\kappa/n = 0.18$: operating costs are about 18% of mark-to-market net worth per year

- $\kappa(n)$ as an *operating cost* of the franchise: DeMarzo et al. (2025)

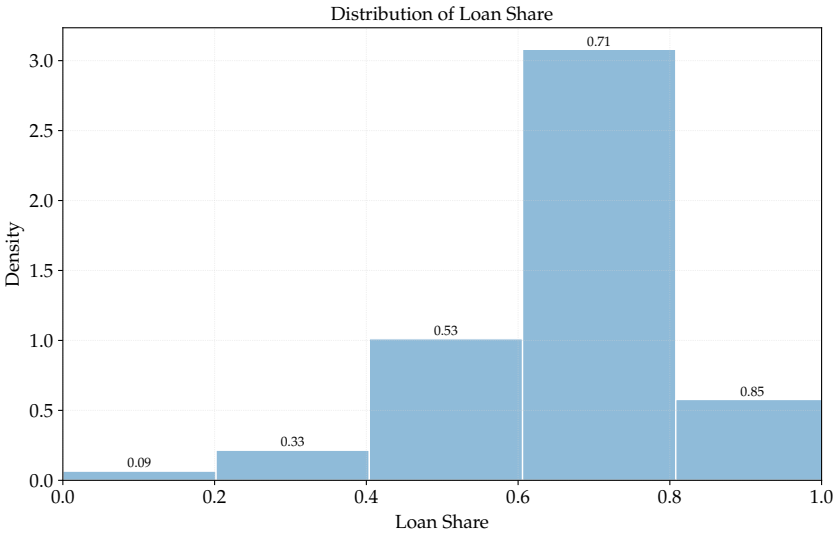
Term Premium

◂ Back ◂ Calibration



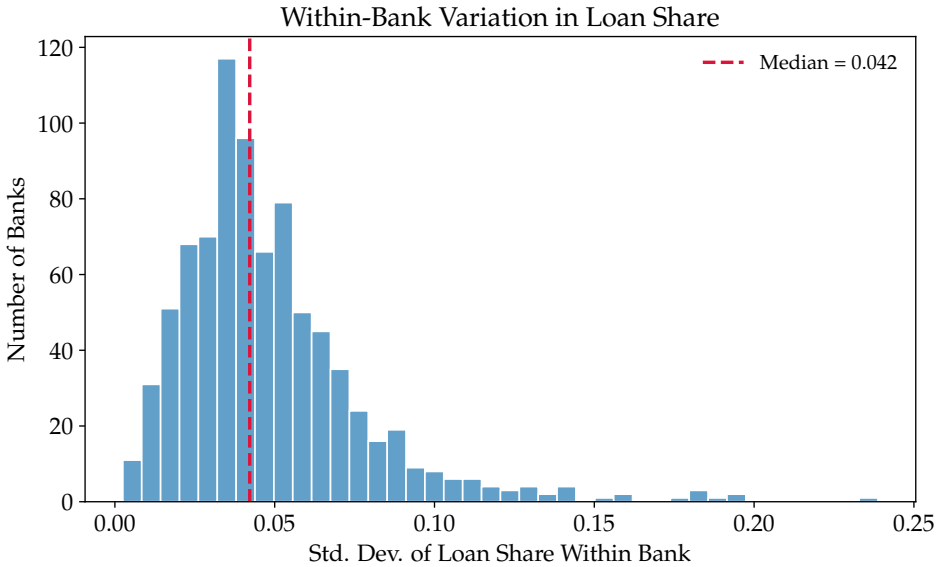
Loan shares I

Back



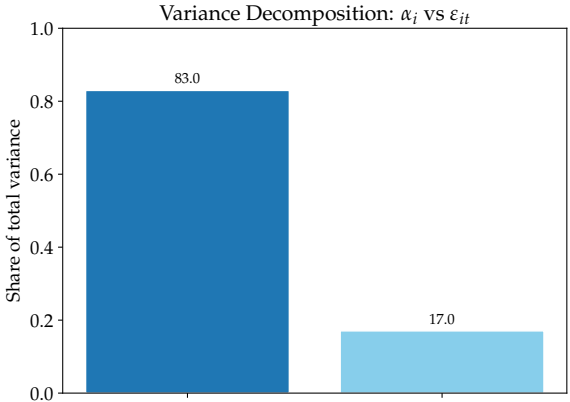
Loan shares II

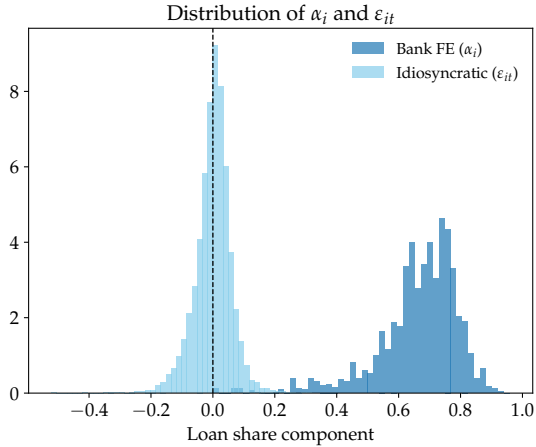
Back



Var. decomposition: $\ell_{it} = \alpha_i + \varepsilon_{it}$

Back





Equilibrium

◂ Back

Equilibrium Definition

1. $\{v, c, \alpha\}$ solve the HJB (2)
2. The taxes τ balance the FDIC's budget (1)
3. The distribution g evolves according to the KFE (3), and banks' forecasts are consistent with its true law of motion

- The distribution g is a **state variable**
  $\Rightarrow$ we solve the model following Schaab (2020)

▸ Algorithm

HJB Equation

Back

$$\rho v(n,x,\ell,r,g) = \max_{c,\alpha} \Big\{ u(c) + v_n \mu_n(n,d,\ell,r,g;\alpha,c) + v_x \bar{x} + v_r \theta_r(\bar{r}-r) + \frac{1}{2}\sigma_n^2(n,x,\ell;\alpha)\, v_{nn} + \frac{1}{2}\sigma_x^2(x)\, v_{xx} + \frac{1}{2}\sigma_r^2\, v_{rr} - \sigma_n(n,x,\ell;\alpha)\sigma_r r\, v_{nr} + \lambda(n,x,\ell)\Big(\bar{v}(n,r) - v(n,x,\ell,r,g)\Big) + \int \frac{\partial v}{\partial g} \times \dot{g}\, dn dx d\ell \Big\}. \tag{2}$$

Distribution dynamics

◂ Back ◂ Equilibrium

$$
\begin{aligned}
dg_t(n,x,\ell) = \Bigg[& -\frac{\partial}{\partial n}\Big(\mu_n(n,x,\ell,r_t,g_t)g_t(n,x)\Big) - \frac{\partial}{\partial x}\Big(\bar{x}g_t(n,x)\Big) \\
& + \frac{1}{2}\frac{\partial^2}{\partial n^2}\left(\sigma_n(n,x,r_t,g_t)^2 g_t(n,x)\right) + \frac{1}{2}\frac{\partial^2}{\partial x^2}\left(\sigma_x(x)^2 g_t(n,x)\right) \\
& - \underbrace{\lambda\Big(n,x,\ell,r_t\Big)g_t(n,x)}_{\text{Exit from } (n,x,\ell) \text{ due runs}} + \underbrace{\mathcal{E}_t(n,x)}_{\text{Entrants in } (n,x,\ell)} \Bigg] dt \\
& - \frac{\partial}{\partial n}\Big(\sigma_n(n,x,\ell,r_t,g_t)\sigma_r g_t(n,x,\ell)\Big) dB_t.
\end{aligned} \tag{3}
$$

Where

$$
\mathcal{E}_t(n,x,\ell) = \underbrace{\psi_0(n,x,\ell)}_{\text{distr. of entrants}} \times \underbrace{\int \lambda(\tilde{n},\tilde{x},\tilde{\ell}) g_t(\tilde{n},\tilde{x},\tilde{\ell})\, d\tilde{n}\, d\tilde{x} d\tilde{\ell}.}_{\text{mass of runs (failures) at } t}
$$

Depositors

- Each bank has a continuum of atomistic, **risk-neutral** depositors indexed by h
- While the bank operates normally ($J_{it} = 0$), deposits follow

$$dx_{it} = (1 - J_{it-})\left[\bar{x}\, dt + \sigma_x\, dW^x_{it} - x_{it-}\, dJ_{it}\right]$$

  baseline: $\bar{x} = \sigma_x = 0$, deposits are constant until a run

- **Coordination opportunities** arrive at Poisson rate $\lambda_0 > 0$
  - The balance sheet is fixed at assets $n + x$
  - Depositors simultaneously choose whether to **request withdrawal** or **stay**
  - $\mathcal{W} \in [0, 1]$: fraction requesting withdrawal

$\Longrightarrow$ the run intensity of J_{it} is derived from this coordination game

Information

- At an opportunity, depositor h observes a **private signal** about bank i:

$$s_{hit} = z_{it} + \sigma_\varepsilon \varepsilon_{hit}, \qquad \varepsilon_{hit} \sim U[-1, 1] \text{ independent}$$

- The **common state** is

$$z_{it} = \frac{n_{it}}{x_{it}} + \sigma_\eta \eta_{it}, \qquad \sigma_\eta > 0$$

  - η_{it}: transitory disturbance to the bank's ability to raise funds, redrawn at each opportunity
  - Depositors do not observe z, but know its prior density f_z
  - The regulator observes z when applying the closure rule, after requests are submitted

Failure

- Securities sell at par; loans sell at a fire-sale price $\delta \in (0,1)$ once the shock has passed
- At the opportunity, the shock also affects sale proceeds. **Liquidation capacity**:

$$x\,[1-\ell(1-\delta)](1+z), \qquad \text{at } \eta = 0: \;\; (1-\ell)(n+x) + \delta\ell(n+x)$$

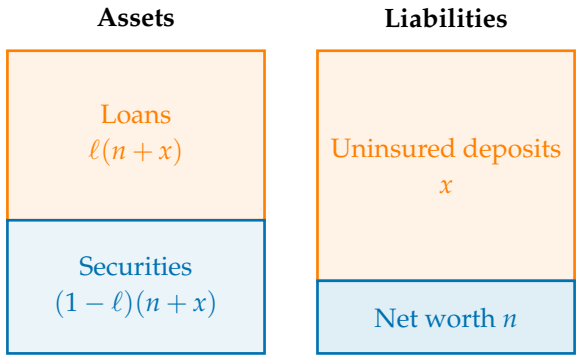
- Requests are recorded before any payment. The bank **fails** if and only if

$$z < \underline{z} \quad \text{or} \quad \mathcal{W} > [1-\ell(1-\delta)](1+z)$$

  - $\underline{z}$: net worth floor below which the regulator closes the bank regardless of requests
  - Upon failure, convertibility is suspended: no request is executed before resolution

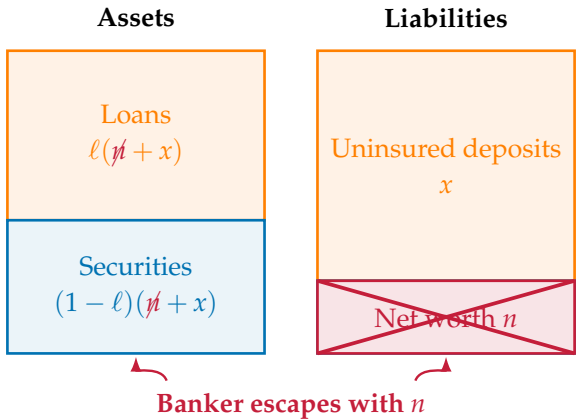
Failure and liquidation

◂ Deposits and runs



Failure and liquidation

Requests exceed liquidation capacity: $\mathcal{W} > [1 - \ell(1-\delta)](1+z) \Rightarrow$ the bank fails



Failure and liquidation

Remaining assets

$$\ell x\,\delta + (1-\ell)x \;=\; [1-\ell(1-\delta)]x$$

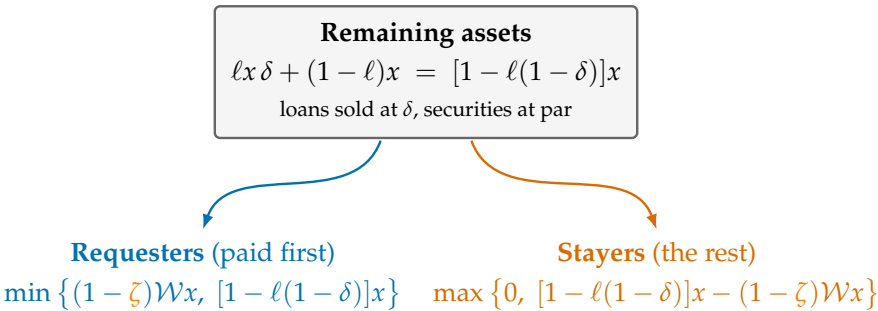
loans sold at δ, securities at par

Requesters (paid first)

$$\min\left\{(1-\zeta)\mathcal{W}x,\; [1-\ell(1-\delta)]x\right\}$$

On top, the insurance fund pays every depositor ζ per dollar of the original claim

Failure and liquidation



On top, the insurance fund pays every depositor ζ per dollar of the original claim

Resolution and payments

- The banker retains assets worth n; the remaining assets yield $[1-\ell(1-\delta)]x$
- The insurance fund pays each depositor ζ per dollar, requester or stayer, with $0 \le \zeta < \ell(1-\delta)$
- Remaining assets pay **requesters first**, then stayers, pro rata within each group:

$$\mathcal{R}(\mathcal{W}) = \min\left\{1, \frac{1-\ell(1-\delta)}{(1-\zeta)\mathcal{W}}\right\}, \qquad \mathcal{S}(\mathcal{W}) = \frac{\max\{0, 1-\ell(1-\delta)-(1-\zeta)\mathcal{W}\}}{(1-\zeta)(1-\mathcal{W})}$$

- Changing $\mathcal{W}$ only redistributes between requesters and stayers. The liquidation loss is split as

$$\underbrace{[\ell(1-\delta)-\zeta]x}_{\text{depositors' loss}} + \underbrace{\zeta x}_{\text{insurer's net expense}} = \underbrace{\ell(1-\delta)x}_{\text{liquidation loss}}$$

Payoffs

◂ Deposits and runs

- Moving the banking relationship costs a fraction $k \in (0,1)$ of each dollar received: a requester pays it at survival, both groups pay it at failure

$$\frac{U^{\mathcal{S}}(z,\mathcal{W})}{\Delta} = \begin{cases} 1, & \text{survives} \\ (1-k)\{\zeta + (1-\zeta)\mathcal{S}(\mathcal{W})\}, & \text{fails} \end{cases}$$

$$\frac{U^{\mathcal{R}}(z,\mathcal{W})}{\Delta} = \begin{cases} 1-k, & \text{survives} \\ (1-k)\{\zeta + (1-\zeta)\mathcal{R}(\mathcal{W})\}, & \text{fails} \end{cases}$$

- **Withdrawal advantage** $\mathcal{G} \equiv (U^{\mathcal{R}} - U^{\mathcal{S}})/\Delta$:

$$\mathcal{G}(z,\mathcal{W}) = \begin{cases} -k, & \text{bank survives} \\ (1-\zeta)(1-k)[\mathcal{R}(\mathcal{W}) - \mathcal{S}(\mathcal{W})], & \text{bank fails} \end{cases}$$

$\Longrightarrow$ withdraw at failure (requesters have priority), stay at survival

Multiplicity and equilibrium selection

◂ Deposits and runs

- Without uncertainty about the bank's ability to raise funds ($\sigma_\varepsilon = \sigma_\eta = 0$), for

$$\underline{z} \leq z < \frac{\ell(1-\delta)}{1-\ell(1-\delta)}$$

  the bank can continue if depositors stay, but cannot meet withdrawals from all of them

- Expectations of others' requests are **self-fulfilling**: anticipating failure, each depositor requests to obtain priority in resolution, and the requests cause the failure

- If depositors stay, the disturbance passes without changing the balance sheet, lending or deposit relationships

  $\Longrightarrow$ global games: payoff uncertainty and private signals select an equilibrium

  Carlsson and Van Damme (1993), Morris and Shin (1998), Morris and Shin (2003)

  No global strategic complementarity (an extra withdrawal can dilute others' recovery): the proof uses the single-crossing approach of Morris and Shin (2003) and Goldstein and Pauzner (2005)

Depositor threshold equilibrium

Proposition *(Depositor threshold equilibrium)*

Define z^* by

$$\begin{cases} z^* = \underline{z}, & \text{if } \int_0^1 \mathcal{G}(\underline{z}, \mathcal{W})\, d\mathcal{W} \leq 0, \\ \int_0^1 \mathcal{G}(z^*, \mathcal{W})\, d\mathcal{W} = 0, & \text{if } \int_0^1 \mathcal{G}(\underline{z}, \mathcal{W})\, d\mathcal{W} > 0, \end{cases}$$

where the second case has a unique solution above $\underline{z}$. Then:

1. For every $\sigma_\varepsilon > 0$ a symmetric threshold equilibrium exists; every equilibrium signal threshold lies strictly between $\underline{z} - \sigma_\varepsilon$ and $\ell(1-\delta)/[1-\ell(1-\delta)] + \sigma_\varepsilon$.
2. For all sufficiently small positive noise, the equilibrium is unique within symmetric threshold strategies, including when $z^* = \underline{z}$.
3. As $\sigma_\varepsilon \to 0$, equilibrium signal thresholds converge to z^*, and so does the state failure boundary. For fixed $z \neq z^*$, the withdrawal fraction converges to one if $z < z^*$ and to zero if $z > z^*$.
4. $\underline{z} \leq z^* < \ell(1-\delta)/[1-\ell(1-\delta)]$. The cutoff is weakly decreasing in k and ζ; at an interior solution it is strictly decreasing in k, and in ζ when $z^* < \zeta/(1-\zeta)$ (independent of ζ when $z^* \geq \zeta/(1-\zeta)$). As $k \downarrow 0$, it converges to $\ell(1-\delta)/[1-\ell(1-\delta)]$.

Implementing (Schaab 2020)

Back

- We need to approximate g as a finite dimension representation:

$$\hat{g}_t(z) = F(\omega_t)(z) \approx g_t(z)$$

- Our goal then is to solve an approximate economy:

$$v(z, r, g) \approx v(z, r, \omega)$$

- For that, households need to forecast the path ω_t:

$$d\omega_t = \mu_\omega(r_t, \omega_t)dt + \sigma_\omega(r_t, \omega_t)dB_t$$

- Define

$$\omega = \int \lambda(z) x g_t(z) dz$$

$$d\omega = \int \lambda(z) x dg_t(z) dz$$

Lemma

For some parametric family $\hat{g}$, the distribution dynamics is given by

$$\sigma_\omega(r_t, \omega_t) = -\int \lambda(n, x, \ell) x \, \partial_n\left(\sigma_n \hat{g}\right) dn dx d\ell$$

$$\mu_\omega(r_t, \omega_t) = \int \lambda(n, x, \ell) x \Big[-\partial_n\left(\mu_n \hat{g}\right) - \partial_x\left(\mu_x \hat{g}\right) + \frac{1}{2}\partial_{nn}\left(\sigma_n^2 \hat{g}\right) + \frac{1}{2}\partial_{xx}\left(\sigma_x^2 \hat{g}\right) - \lambda \hat{g} + \mathcal{E}(n, x, \ell) \Big] dn dx d\ell$$

Algorithm

Back

1. Pick a reference density $g_0(z)$
2. Choose the moment we want the aggregate state to represent:

$$\omega_0 = \int \lambda(z) x g_0(z)\, dz$$

3. Build the ω-grid

$$\omega_w \in [\omega_0 - h_w, \omega_0 + h_w]$$

4. Compute $\hat{g}(z; \omega_w)$ as the tilted version of g_0

$$\hat{g}(z; \omega_w) = \frac{g_0(z) \exp\{\theta(\omega_w)\lambda(z)x\}}{\int g_0(\tilde{z}) \exp\{\theta(\omega_w)\lambda(\tilde{z})\tilde{x}\} d\tilde{z}}$$

Where $\theta(\omega_w)$ tilts the distribution family so its moment matches the gridpoint for ω_w is such that:

$$\omega_w = \int \lambda(z) x \hat{g}_{\theta(\omega_w)} dz$$

5. Using the distribution $\hat{g}(\omega_w)$ we can compute:

$$\tau(\omega_w) = \int \lambda(z) x \hat{g}(z; \omega_w) dz$$

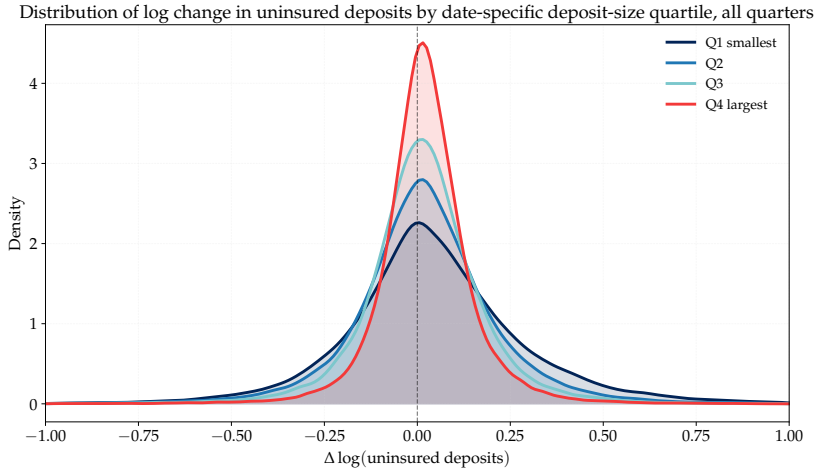
6. With the tax in hand:
   a. Guess a value function v_0 and iterate $v_0 \rightarrow v_1 \rightarrow \cdots \rightarrow v_s$
   b. At iteration s:
      i. Policies for the iteration s: c^s, α^s, μ_n^s, σ_n^s (using μ_ω^{s-1}, and σ_ω^{s-1})
      ii. Use policies at s and the closure $\hat{g}(.; \omega_w)$ to get the dynamics of ω at ω_w:

$$\mu_\omega^s(., \omega_w) \qquad \text{and} \qquad \sigma_\omega^s(., \omega_w)$$

   c. At convergence: $||v^{s+1} - v^s|| < \text{tol}$ we get v^*, c^*, α^*, μ_ω^*, and σ_ω^*.

Quarterly changes in uninsured deposits

Back



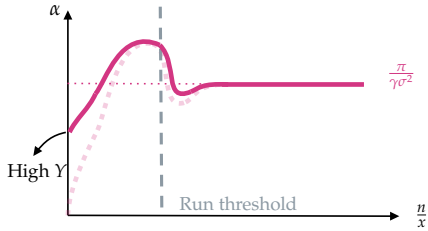
Δ log uninsured deposits between consecutive quarters, by size quartile of the quarter: mean 0.02–0.04, s.d. 0.12–0.13 in every quartile.

Last time: the mechanism came from the **myopic** demand

Portfolio characterization

The portfolio weight of long bonds is

$$\alpha^*(n,x,\ell,r) = \underbrace{\frac{-v_n}{v_{nn}(1-\ell)(n+x)}}_{1/RRA} \times \frac{\pi}{\sigma^2} + \text{Dynamic hedging demand}$$



Welfare of banks: the consumption equivalent

◂ Back

1. **Worlds and rate paths.** Rates from 2010Q1 to $\bar{T} = 2025Q2$: the observed path (Realized) or 1,000 draws of the model's rate process from 2021Q4-2025Q2 (All paths). Banks start at 2010Q1 at $(n_{i0}, x_{i0}, \ell_{i0})$; each follows its own optimal (c, α) and fee (1). Entrants are not counted.

2. **Welfare of bank** i, computed in each world (T_i: its run date, $s = (n, x, \ell, r)$):

$$W_i = \int_0^{T_i \wedge \bar{T}} e^{-\rho t}\, u(c_{it})\, dt \;+\; \mathbb{1}_{\{T_i \le \bar{T}\}}\, e^{-\rho T_i}\, \bar{v}\big(n_{iT_i}, r_{T_i}\big) \;+\; \mathbb{1}_{\{T_i > \bar{T}\}}\, e^{-\rho \bar{T}}\, v\big(s_{i\bar{T}}\big)$$

3. **Consumption equivalent.** In the benchmark (no FDIC, no floor), scale dividends by $1 + \mu$ at every date and state while the bank operates. The extra is paid from outside, so the balance sheet does not change:

$$W_i(\mu) = \int_0^{T_i \wedge \bar{T}} e^{-\rho t}\, u\big((1+\mu)\, c_{it}\big)\, dt \;+\; \mathbb{1}_{\{T_i \le \bar{T}\}}\, e^{-\rho T_i}\, \bar{v}\big(n_{iT_i}, r_{T_i}\big) \;+\; \mathbb{1}_{\{T_i > \bar{T}\}}\, e^{-\rho \bar{T}}\, v_\mu\big(s_{i\bar{T}}\big)$$

   v_μ keeps the policies: $\big(\rho + \lambda - \mathcal{A}\big)\, u\big((1+\mu)\, c\big) + \lambda\, \bar{v}$, one linear solve per μ.

4. **Aggregate.** One common $\bar{\mu}$ for all banks (All paths: and all 1,000 paths):

$$\sum_i \underbrace{W_i(\bar{\mu})}_{\text{benchmark: no FDIC, no floor}} = \sum_i \underbrace{W_i}_{\text{policy world}} \quad \Rightarrow \quad \text{the table reports } 100\,\bar{\mu}$$

Reading: FDIC, no floor (Realized): $\bar{\mu} = -1.72\% \Rightarrow$ the fee costs the 2010 banks as much as a permanent 1.72% dividend cut.

$Y > 0$ makes u non-homothetic, so $\bar{\mu}$ is not a ratio of values: $W_i(\mu)$ increases in μ, is evaluated on a grid and inverted.

Duration Restrictions: Realized vs. All Paths

Back

	Deposits in run banks		Sec. losses in 2022 (%)		Welfare of 2010 banks (%)	
	Realized	All paths	Realized	All paths	Realized	All paths
FDIC, no floor	1.00	0.92	−17.3	0.0	0.00	0.00
FDIC + floor $\phi = 0.1$	0.99	0.93	−16.6	0.0	0.80	−0.30
FDIC + floor $\phi = 0.2$	0.94	0.94	−11.1	0.0	3.19	−2.52
FDIC + floor $\phi = 0.3$	0.89	0.97	−5.2	0.0	4.46	−4.74
FDIC + floor $\phi = 0.4$	0.87	0.98	−2.3	0.0	4.21	−5.97

All paths: observed rates to 2021Q4, then 1,000 draws of the model's rate process to 2025Q2. Deposits relative to Realized, "FDIC, no floor"; welfare relative to FDIC, no floor within each column.

Entrants and incumbents

Back

Table: Distribution of n/x: incumbents vs. entrants

Quantile	Incumbents	Entrants
5%	0.17	0.16
25%	0.29	0.28
50%	0.42	0.45
75%	0.58	0.65
95%	0.96	1.51

Capital-floor experiment: interpretation

Back

- The floor operates before the shock: banks enter 2022 with different portfolios
- The remaining stress scenario declines over the episode as an unanticipated release
- The floor evaluates current exposure; endogenous balance-sheet changes mean it is not a guarantee on realized losses
- Net-worth changes use each policy's own pre-shock base; they do not compare final net-worth levels across policies
- A shock chosen to match securities losses need not match the observed rate path

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